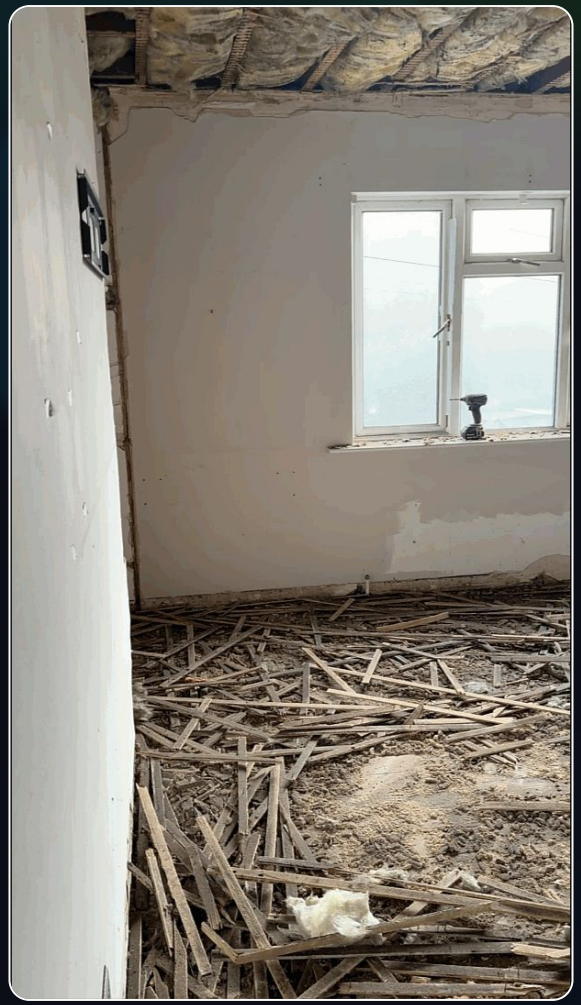


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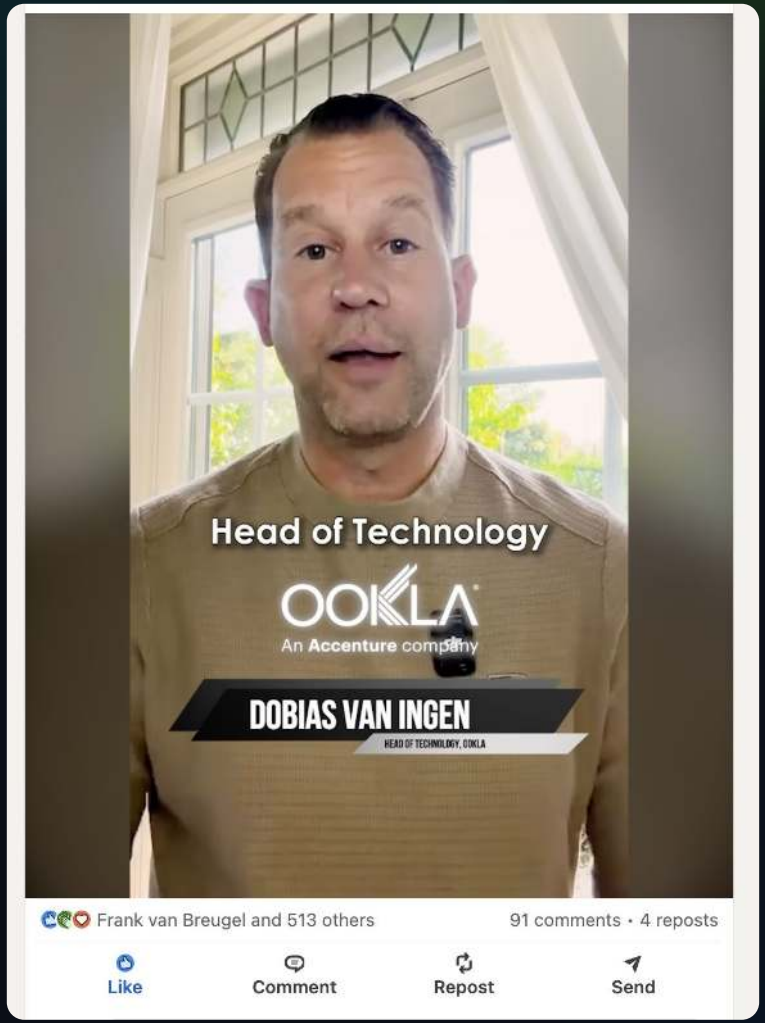


Proprietary & Confidential



Ookla to become **Part of Accenture**




Head of Technology
OOKLA
An Accenture company
DOBIAS VAN INGEN
HEAD OF TECHNOLOGY, OOKLA
Frank van Breugel and 513 others
91 comments • 4 reposts
Like
Comment
Repost
Send

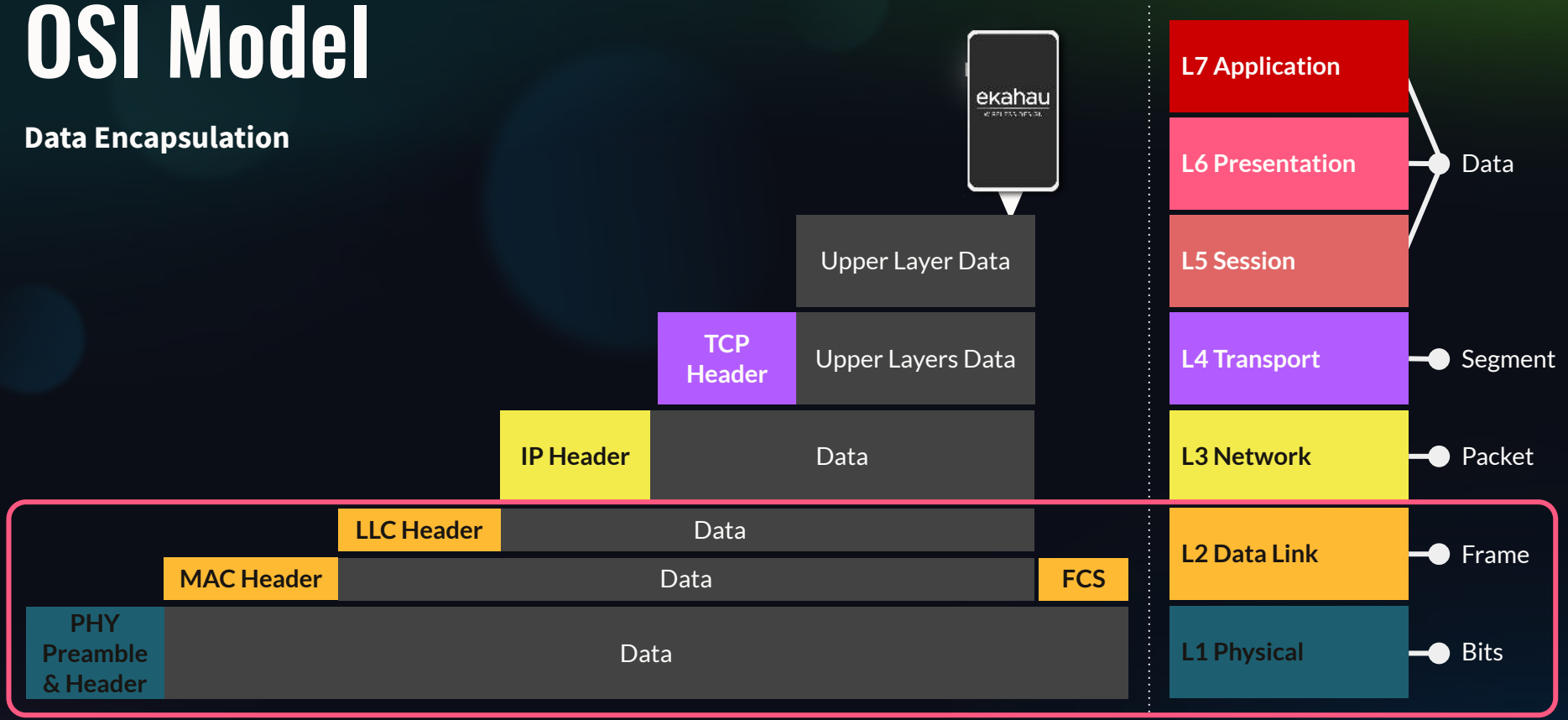
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I  

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OSI Model

Data Encapsulation



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WIRELESS DIAGNOSTICS & TELEMETRY

Wi-Fi Architecture: Layer 1 vs. Layer 2

Distinguishing raw electromagnetic RF energy from 802.11 protocol frame metrics.

PHYSICAL MEDIUM (PHY)

Layer 1: RF Energy



Raw RF Energy: Unmodulated electromagnetic wave radiation.

Non-Wi-Fi Noise: Bluetooth devices, microwave ovens, radars.

Duty Cycle: Percentage of raw channel saturation.

Diagnostics Tool: Dedicated spectrum analyzer hardware.

DATA LINK (802.11 MAC)

Layer 2: Protocol Frames



Signal Metrics: RSSI decibels, noise floor, SNR margins.

Contention: Co-channel interference (CCI), retry rates.

Data Rates: Negotiated MCS index, PHY link throughput.

Diagnostics Tool: Standard Wi-Fi client NICs, packet captures.

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WI-FI NERDS

SIT FORWARD

MARKETING / SALES

SIT BACK

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How Does Wi-Fi Work?

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THE WI-FI GAME: HOW DEVICES TAKE TURNS TALKING

Clear Channel Assessment (CCA) a.k.a. "The Game" – Avoid Collisions, Get Acked, Win the Airtime

1 LISTEN BEFORE YOU TALK (CCA)

A. PHYSICAL CARRIER SENSE

Check the air for activity

Preamble Detect (Wi-Fi) detects other Wi-Fi devices
- BUSY Wait
- CLEAR Proceed

Energy Detect (Non-Wi-Fi) detects interference & other signals
- BUSY Wait
- CLEAR Proceed

B. VIRTUAL CARRIER SENSE (NAV)

Check if airtime is reserved



Devices hear a transmission and set their Network Allocation Vector (NAV) – they wait for the Duration to expire.

2 THE GAME: CONTENTION WINDOW

Everyone ready to talk rolls a dice (random backoff). Smaller number wins!



STA 3 wins the airtime!

Contention Window (CW) = range of slots to roll from
More busy air = larger CW) = more waiting

3 TRANSMIT OPPORTUNITY (TxOP)

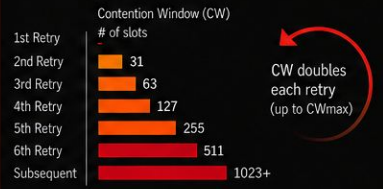
Winner reaches 0 and gets to transmit!



- ACK RECEIVED – Success! Move on.
- NO ACK – Collision or interference. Didn't get it. Retry!

4 RETRY & BACKOFF

If no ACK, the device tries again with a bigger window.



5 NOT ALL TRAFFIC WAITS THE SAME

EDCA / Wi-Fi QoS gives different traffic different priority.

ACCESS CATEGORY	EXAMPLE	CW RANGE
AC_VO (Voice)	Voice	CW 0 – 3
AC_VI (Video)	Video	CW 0 – 7
AC_BE (Best Effort)	Data	CW 0 – 15
AC_BK (Background)	Background	CW 0 – 15

HIGH PRIORITY – Smaller CW = faster access
LOW PRIORITY – Larger CW = more waiting

KEY TAKEAWAY

Wi-Fi tries to avoid collisions by listening, waiting, and taking turns. Sometimes collisions still happen (same roll, interference, etc.), but this is how Wi-Fi works and tries to behave.

More Devices = More Waiting = Less Throughput

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Listen Before You Talk - Clear Channel Assessment

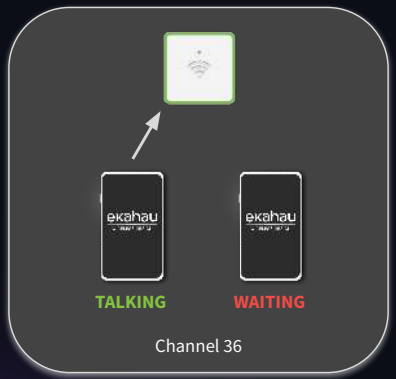
Physical Carrier Sense

Listen before you talk

Preamble Detect (PD) / Signal Detect (SD)

4dB above noise floor

Is there another **Wi-Fi device** transmitting on the channel?



Energy Detect (ED)

20dB above PD

Is there a non-Wi-Fi device transmitting on the channel?

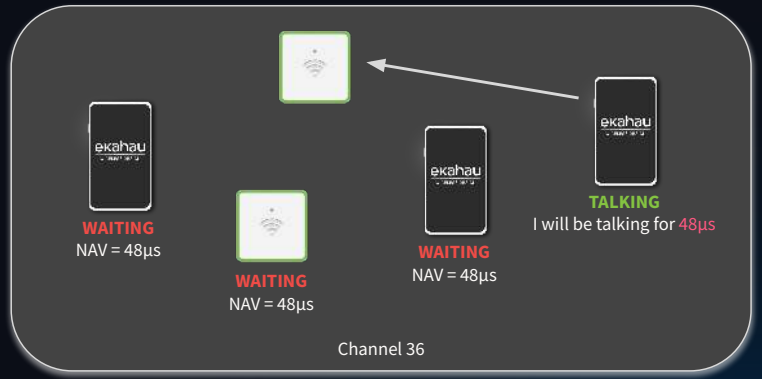


Virtual Carrier Sense

Network reservation

Network Allocation Vector (NAV)

Clients hear a transmission
Clients read the Duration included in every frame
Clients set their NAV timer = to the Duration



Technology Overview

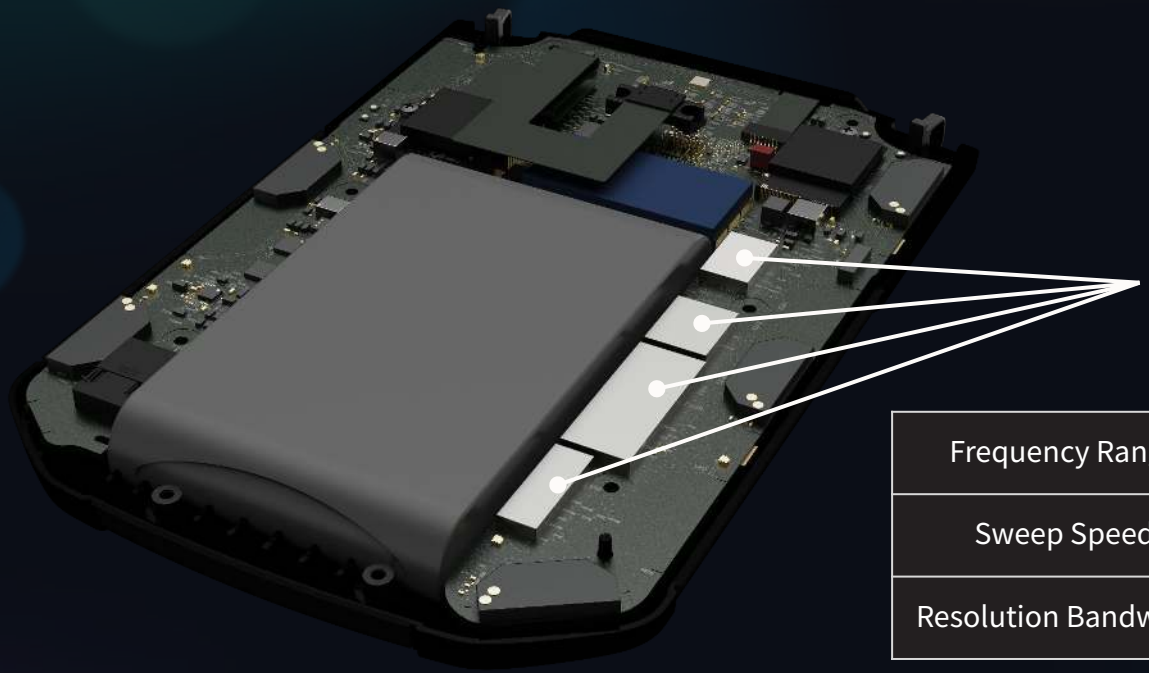
How Does the Sidekick 2 Spectrum Analyzer Work?



A deep dive into high-resolution, multi-band RF spectrum analysis.

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Sidekick 2 Spectrum Analyzer - Key Specs



Frequency Range	**2,400** MHz - **7,125** MHz
Sweep Speed	**50** sweeps/sec
Resolution Bandwidth	≃ **19** KHz

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SK2 Spectrum Analyzer - How Does it Work?

1. RF Sweep (2.4 / 5 / 6 GHz)
2. Down Conversion
3. Digitization (ADC)
4. FFT (Frequency vs. Power)



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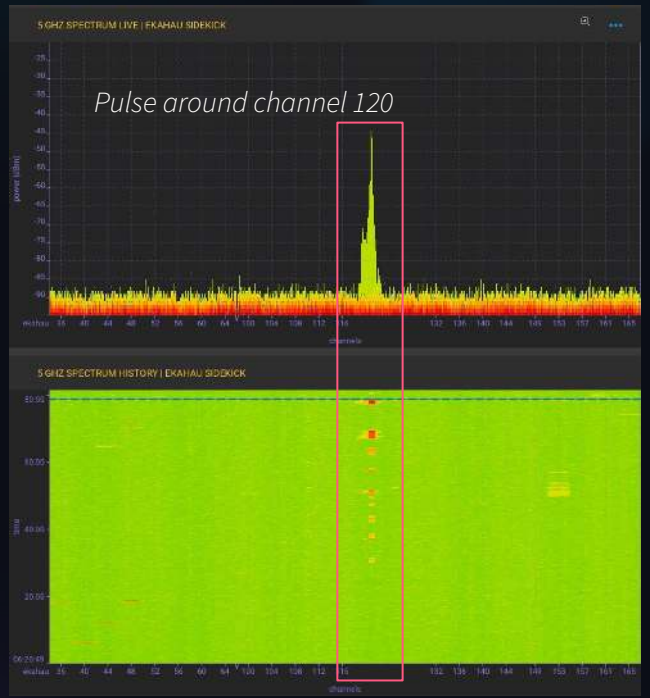
Interference Detection
Signal Fingerprinting

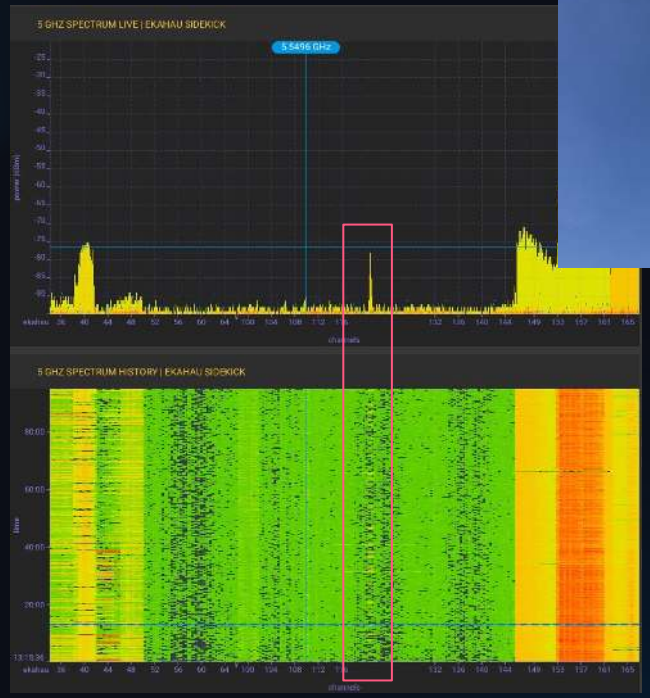
- Ekahau looks at each signal characteristics.
- Classification done based on signal fingerprinting.
- Detection is based on statistical matching (not 100% foolproof).



Weather Radar Pulse

- Short bursts
- Not always easy to see





TDWR Tower



Real World Example - LED Lights / Motion Sensors

- Multiple narrow band signals
- Continuous transmissions around UNII-3 band (5 GHz)





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Real World Example - Parking Gate

- Continuous Wave RADAR
- Continuous transmissions around channel 9 on 2.4 GHz





Real World Example - Live Video Transmitter

- Continuous 20 MHz transmission on 5 GHz or 6 GHz
- Transmitters attached to cameras and moving around

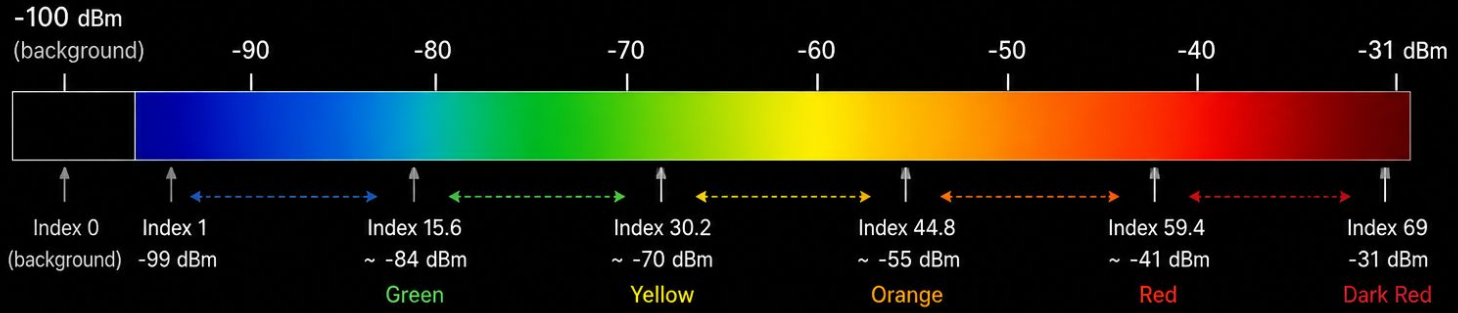




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Spectrum Waterfall Color Legend

Power (dBm) → Color



Power (dBm)	-100	-90	-80	-70	-60	-50	-40	-31
Color Index	0	10	20	30	40	50	60	69
Color (approx.)								
Description	Background (no data)	Very Weak (Noise Floor)	Weak	Moderate	Moderately Strong	Strong	Very Strong	Extremely Strong

- Power range: -100 dBm (background) to -31 dBm (strongest displayed)
- Colors are mapped in 1 dB steps: `index = floor(dBm) + 100`
- Index 0 is background. Index 1 corresponds to -99 dBm. Index 69 corresponds to -31 dBm.
- Full color range (to dark red) extends to -26 dBm (index 74), but the display is configured up to -31 dBm.

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“You Don’t need a Spectrum Analyzer Anymore APs can Detect and Avoid Interference...”

The Promise

Access Points use automated Radio Resource Management (RRM) to optimize channels.

- Continuous environment scanning
- Dynamic channel & power adjustments
- Automatic Wi-Fi congestion avoidance

The Reality

Sure, but what about when they don’t?

- **Blind to Non-Wi-Fi:** Most APs cannot detect continuous raw RF transmitters (microwave, radar).
- **Silent Disruptions:** Video transmitters & motion sensors pollute bands without APs realizing.
- **False Noise Floors:** APs lack true spectrum hardware to measure actual environment noise.

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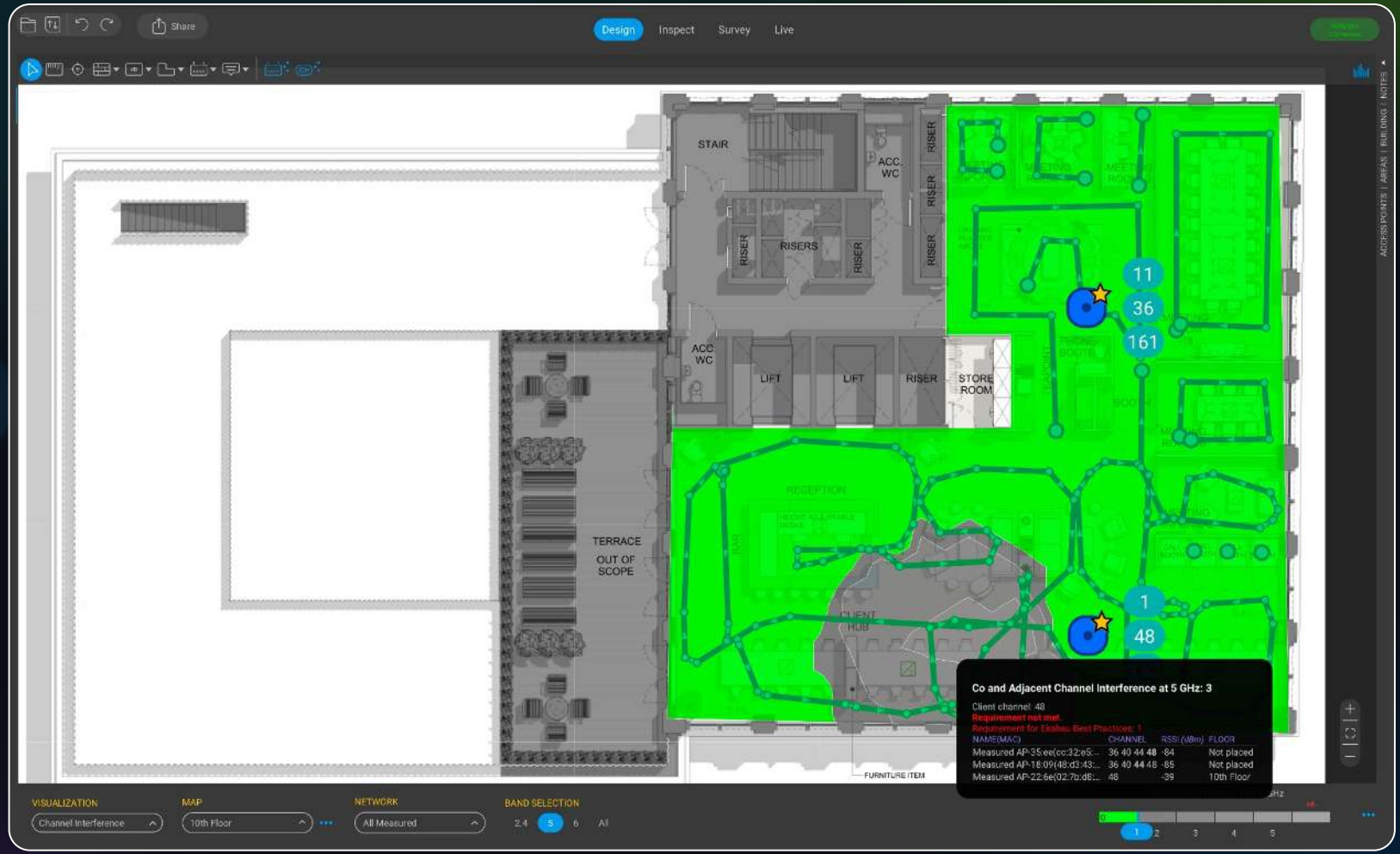
Customer Experiencing Intermittent Wi-Fi Performance

Office Example 1

Proprietary & Confidential


Share
Design
Inspect
Survey
Live
STAIR
ACC WC
RISER
RISERS
LIFT
STORE ROOM
11
36
161
1
48
153
RECEPTION
BAR
TERRACE
OUT OF SCOPE
FURNITURE ITEM
VISUALIZATION
Signal Strength
MAP
10th Floor
NETWORK
All Measured
BAND SELECTION
2.4
5
6
All
Ekahau Best Practices for 5 GHz (dBm)


Share
Design
Inspect
Survey
Live
STAIR
ACC. WC
RISER
RISERS
LIFT
STORE ROOM
TERRACE
OUT OF SCOPE
RECEPTION
MEETING ROOM
11
36
161
1
48
153
VISUALIZATION
Secondary Signal Str...
MAP
10th Floor
NETWORK
All Measured
BAND SELECTION
2.4
5
6
All
Ekahau Best Practices for 5 GHz (dBm)
-67
-60
-55
-50
-45
-40
-35
FURNITURE ITEM

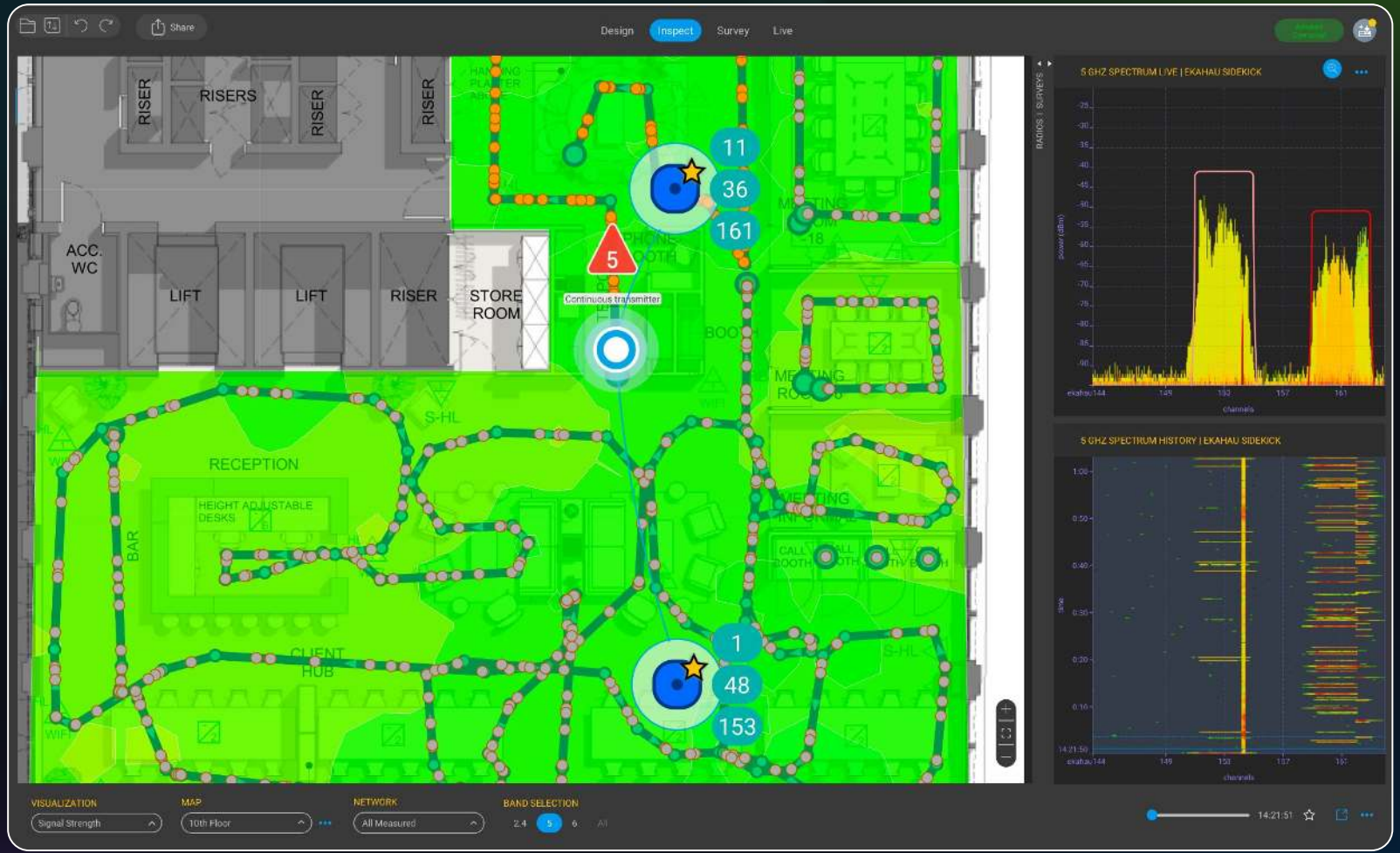

Share
Design
Inspect
Survey
Live
STAIR
ACC. WC
RISER
RISERS
ACC WC
LIFT
LIFT
RISER
STORE ROOM
TERRACE
OUT OF SCOPE
RECEPTION
CLIENT HUB
FURNITURE ITEM
11
36
161
1
48
Co and Adjacent Channel Interference at 5 GHz: 3
Client channel: 48
Requirement not met.
Requirement for Ekahau Best Practices: 1
NAME(MAC) CHANNEL RSSI (dBm) FLOOR
Measured AP-35:ee(cc:32:e5:... 36 40 44 48 -84 Not placed
Measured AP-18:09(48:d3:43:... 36 40 44 48 -85 Not placed
Measured AP-22:6e(02:7b:d8:... 48 -39 10th Floor
VISUALIZATION
Channel Interference
MAP
10th Floor
NETWORK
All Measured
BAND SELECTION
2.4
5
6
All

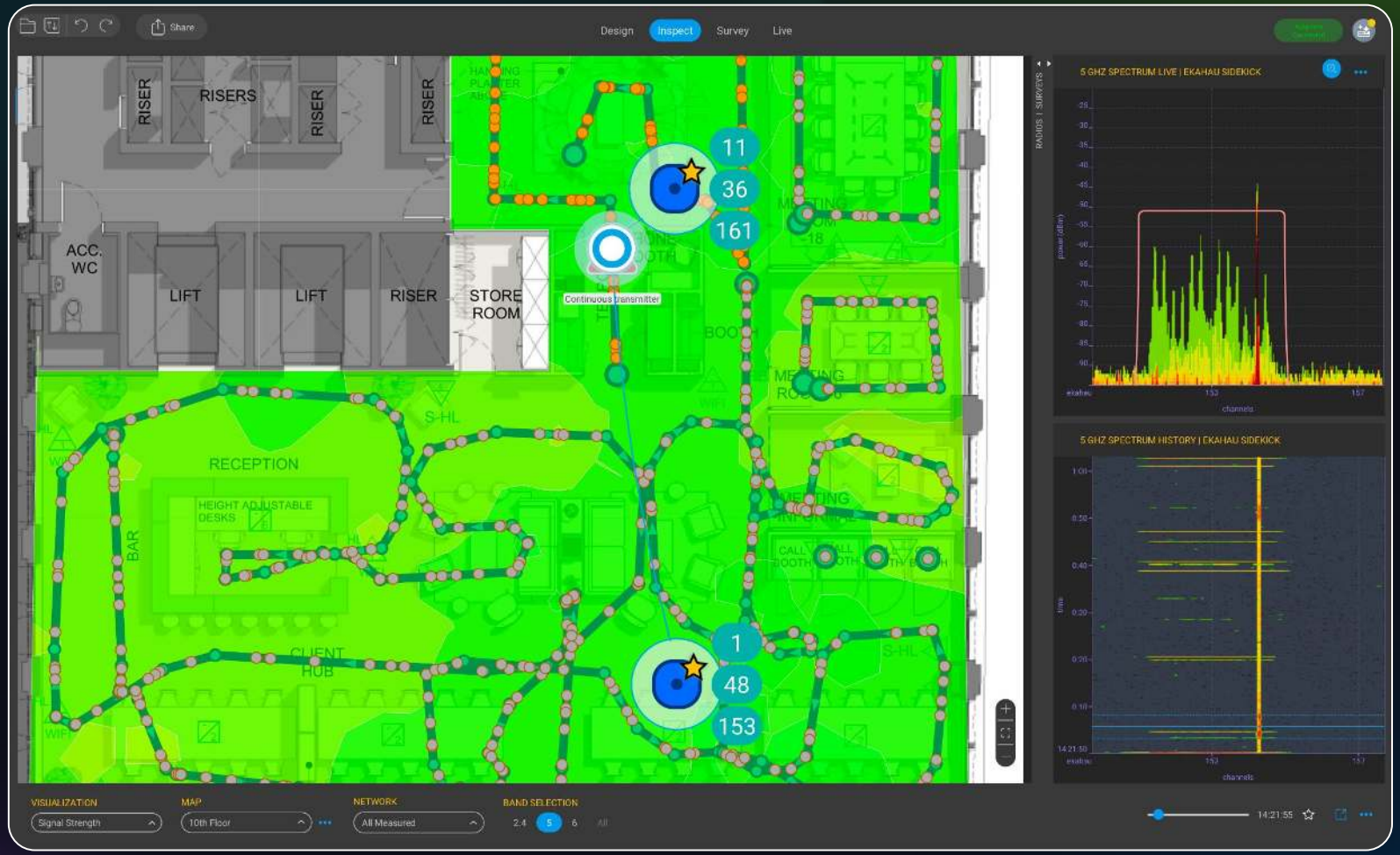
This is all you would see



Without a Spectrum Analyzer

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Share
Design
Inspect
Survey
Live
RISER
RISERS
RISER
RISER
ACC. WC
LIFT
LIFT
RISER
STORE ROOM
11
36
161
5
Continuous transmitter
S-HL
RECEPTION
HEIGHT ADJUSTABLE DESKS
BAR
CLIENT HUB
1
48
153
5 GHZ SPECTRUM LIVE | EKAHAU SIDEKICK
5 GHZ SPECTRUM HISTORY | EKAHAU SIDEKICK
VISUALIZATION
Signal Strength
MAP
10th Floor
NETWORK
All Measured
BAND SELECTION
2.4
5
6
All
14:21:51


Share
Design
Inspect
Survey
Live
RISERS
RISER
RISER
RISER
ACC. WC
LIFT
LIFT
RISER
STORE ROOM
Continuous transmitter
11
36
161
RECEPTION
HEIGHT ADJUSTABLE DESKS
BAR
CLIENT HUB
1
48
153
5 GHZ SPECTRUM LIVE | EKAHAU SIDEKICK
5 GHZ SPECTRUM HISTORY | EKAHAU SIDEKICK
VISUALIZATION
Signal Strength
MAP
10th Floor
NETWORK
All Measured
BAND SELECTION
2.4
5
6
All

Customer Experiencing Intermittent Wi-Fi Performance

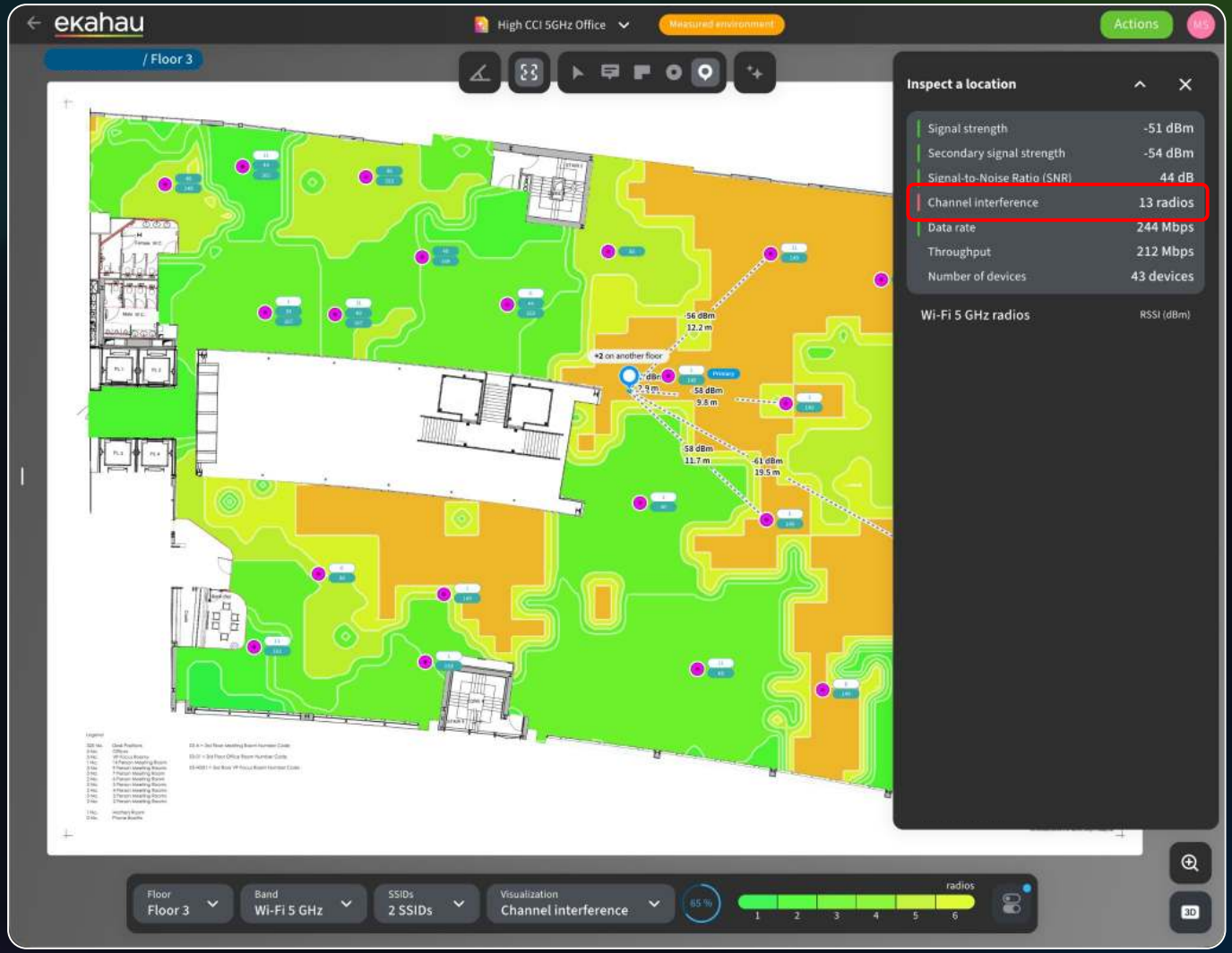
Office Example 2

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ekahau
Office
Measured environment
Actions
/ Floor 3
CONSTRUCTION
3RD FLOOR
Floor
Floor 3
Band
Wi-Fi 5 GHz
SSIDs
2 SSIDs
Visualization
Signal strength
100 %
dBm
-87 -78 -67 -54 -45 -36 -27
3D



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ekahau
High CCI 5GHz Office
Measured environment
Actions
/ Floor 3
Inspect a location
Signal strength -51 dBm
Secondary signal strength -54 dBm
Signal-to-Noise Ratio (SNR) 44 dB
Channel interference 13 radios
Data rate 244 Mbps
Throughput 212 Mbps
Number of devices 43 devices
Wi-Fi 5 GHz radios
RSSI (dBm)
-56 dBm
12.2 m
+2 on another floor
-55 dBm
9.8 m
-58 dBm
11.7 m
-61 dBm
19.5 m
Floor
Floor 3
Band
Wi-Fi 5 GHz
SSIDs
2 SSIDs
Visualization
Channel interference
radios
1 2 3 4 5 6
3D

Any ideas what's the cause of high CCI here?

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Click on the map to inspect the selected point for mor

Frequency Monitor

Radios | Surveys

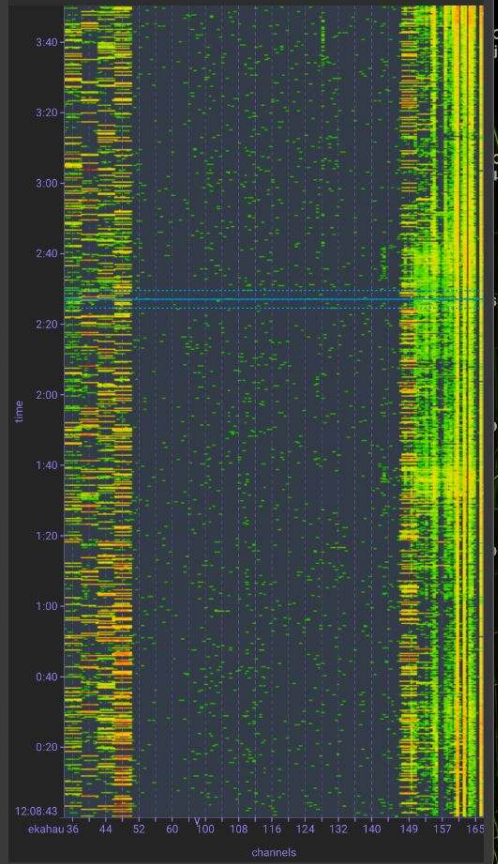
RADIOS (11)

AP NAME	TECH	CH	SNR	RSSI
		149 (20)	45	-46
		149 (20)	42	-49
		149 (20)	41	-50
		149 (20)	40	-51
		149 (20)	38	-53
		149 (20)	34	-57
		149 (20)	34	-59
		149 (20)	32	-60
		149 (20)	23	-68
		149 (20)	18	-73
		149 (20)	-	-

5 GHZ SPECTRUM LIVE | EKAHAU SIDEKICK



5 GHZ SPECTRUM HISTORY | EKAHAU SIDEKICK

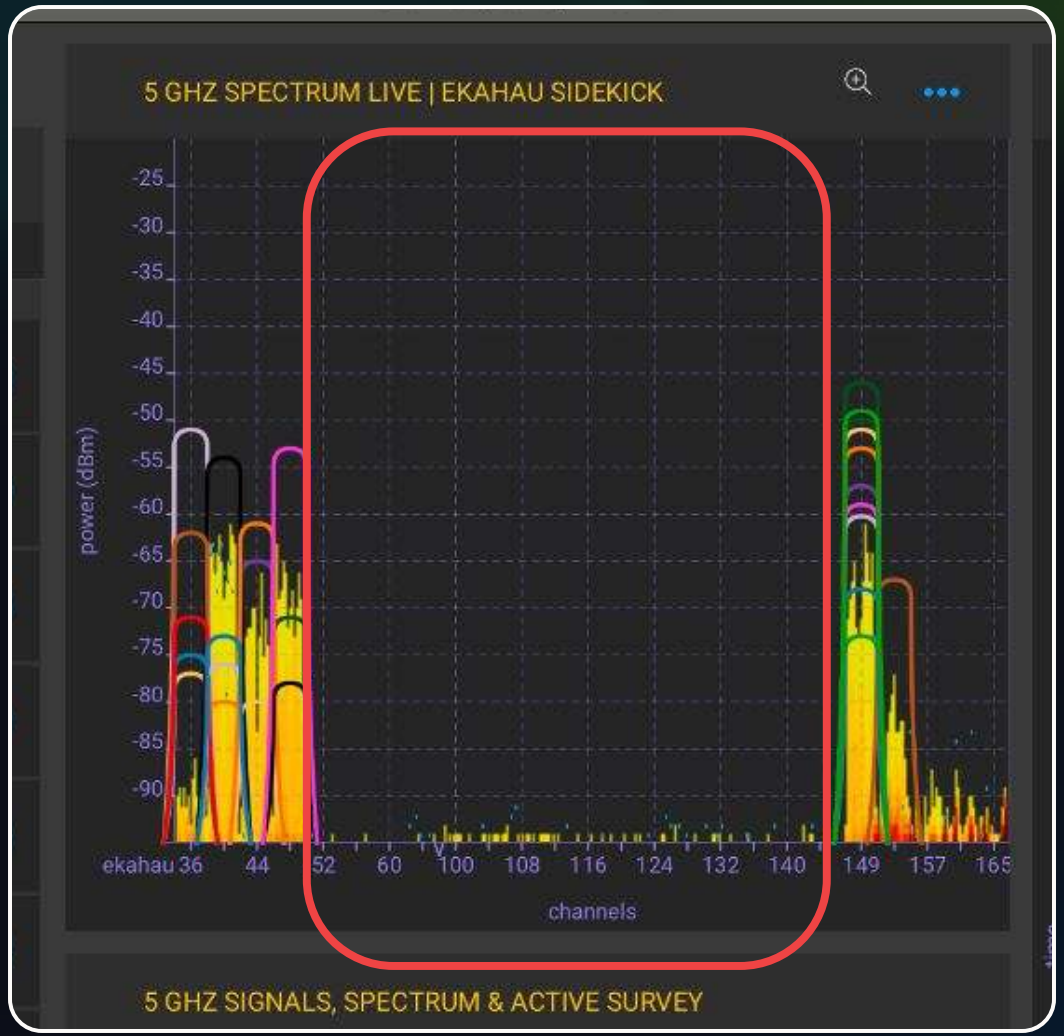


5 GHZ SIGNALS, SPECTRUM & ACTIVE SURVEY



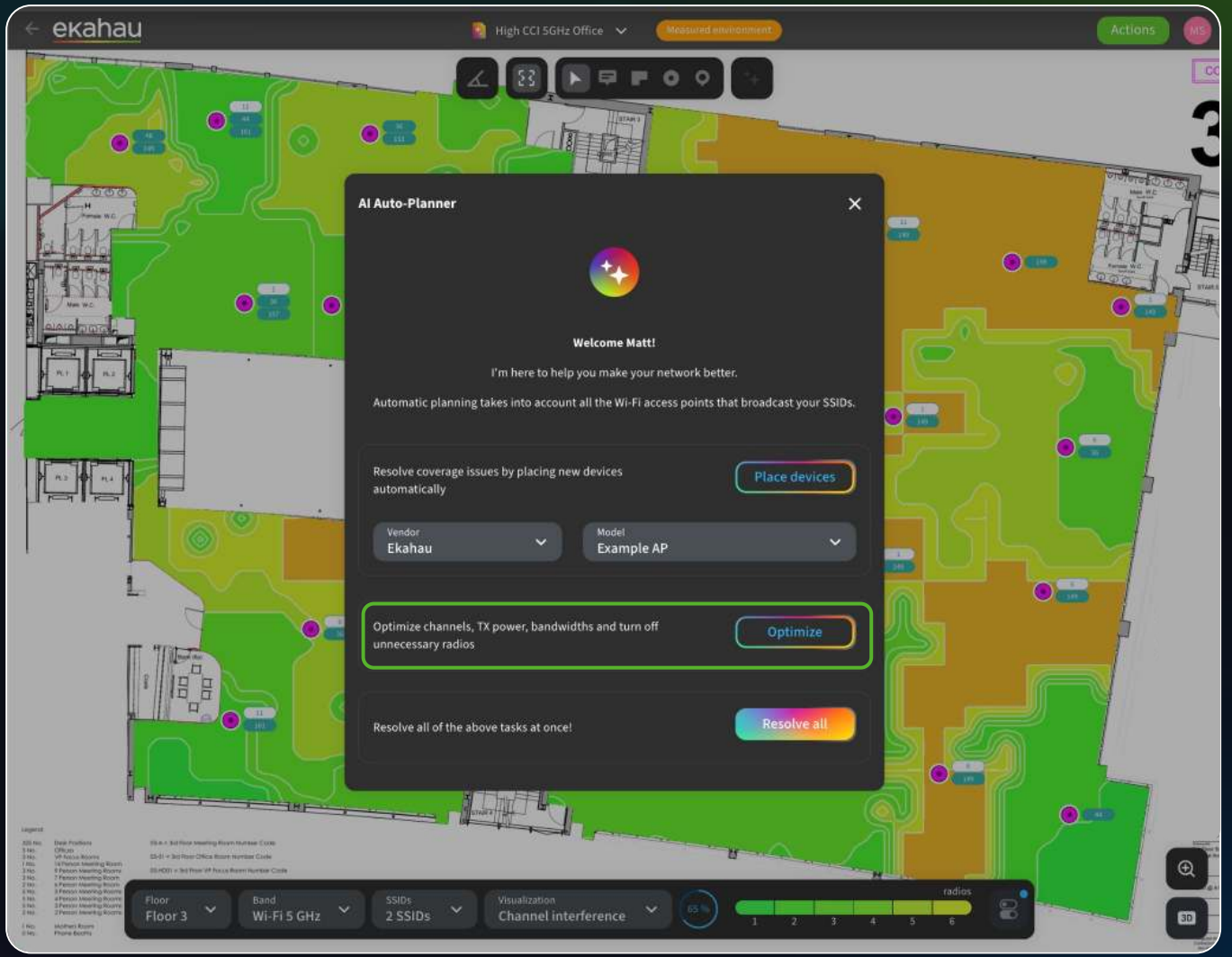
VISUALIZATION | MAP | NETWORK | BAND SELECTION

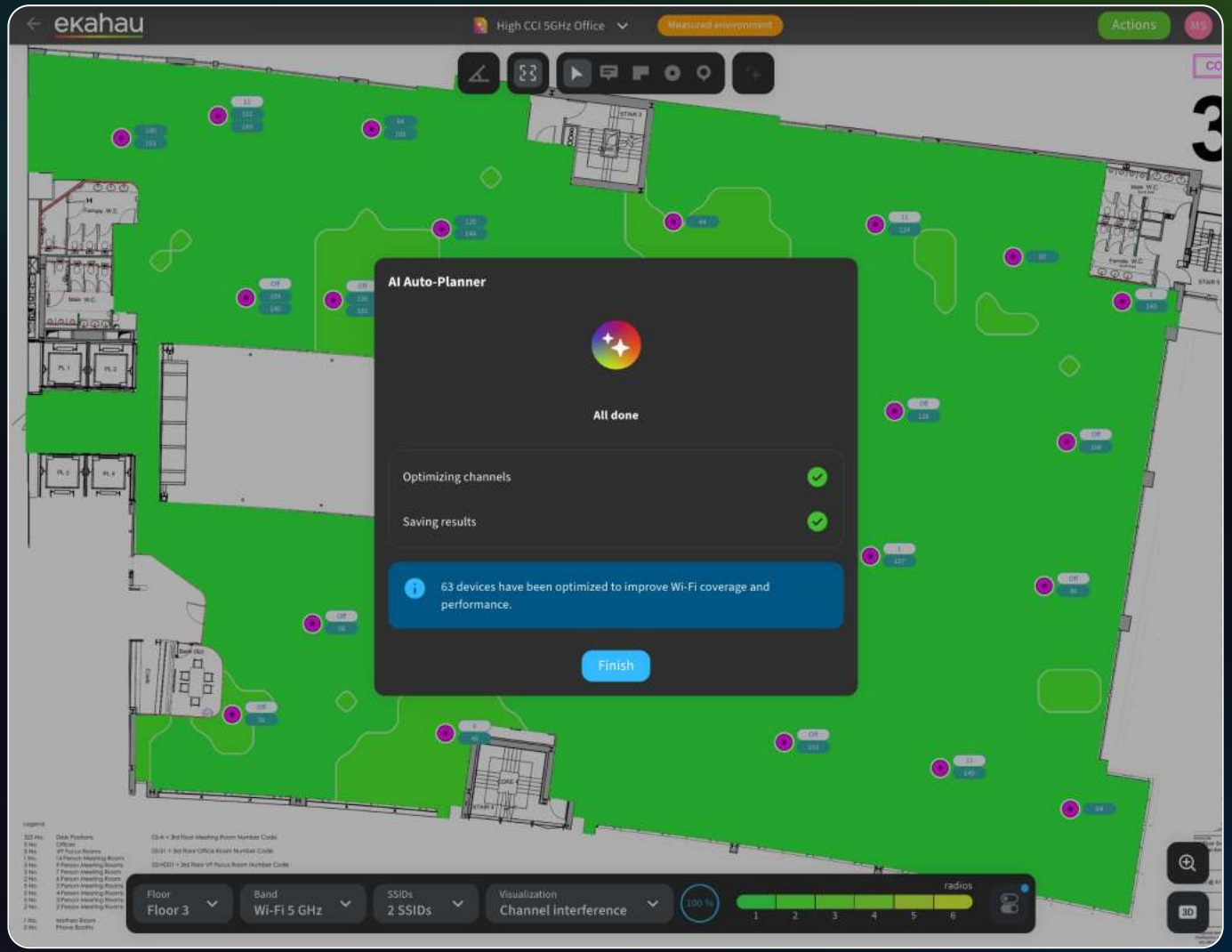
Screen Recording


5 GHZ SPECTRUM LIVE | EKAHAU SIDEKICK
power (dBm)
-25
-30
-35
-40
-45
-50
-55
-60
-65
-70
-75
-80
-85
-90
ekahau 36
44
52
60
100
108
116
124
132
140
149
157
165
channels
5 GHZ SIGNALS, SPECTRUM & ACTIVE SURVEY

OOKLA

Disabled use of DFS channels

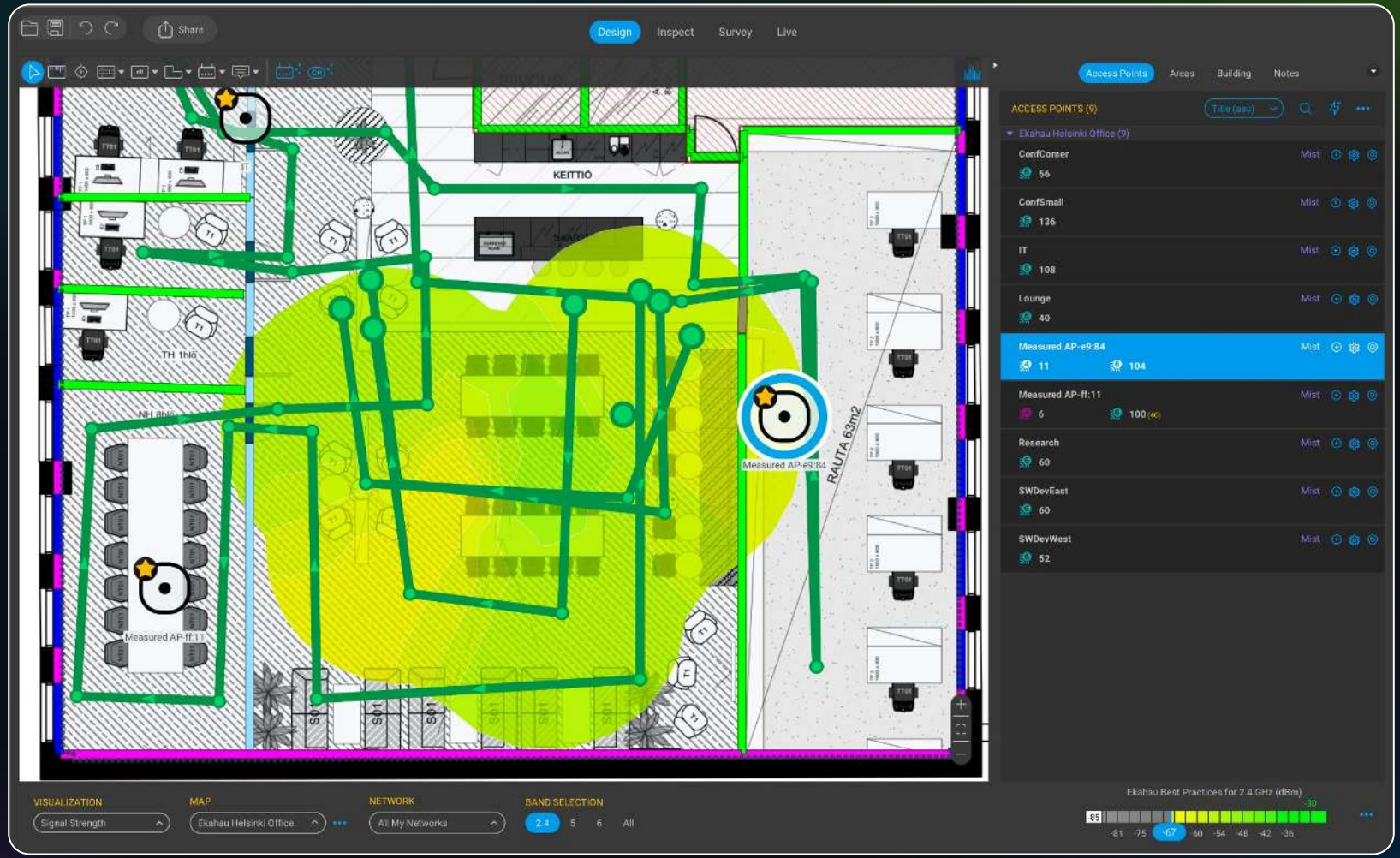

ekahau
High CCI 5GHz Office
Measured environment
Actions
AI Auto-Planner
Welcome Matt!
I'm here to help you make your network better.
Automatic planning takes into account all the Wi-Fi access points that broadcast your SSIDs.
Resolve coverage issues by placing new devices automatically
Place devices
Vendor
Ekahau
Model
Example AP
Optimize channels, TX power, bandwidths and turn off unnecessary radios
Optimize
Resolve all of the above tasks at once!
Resolve all
Floor
Floor 3
Band
Wi-Fi 5 GHz
SSIDs
2 SSIDs
Visualization
Channel interference
radios
1
2
3
4
5
6
3D

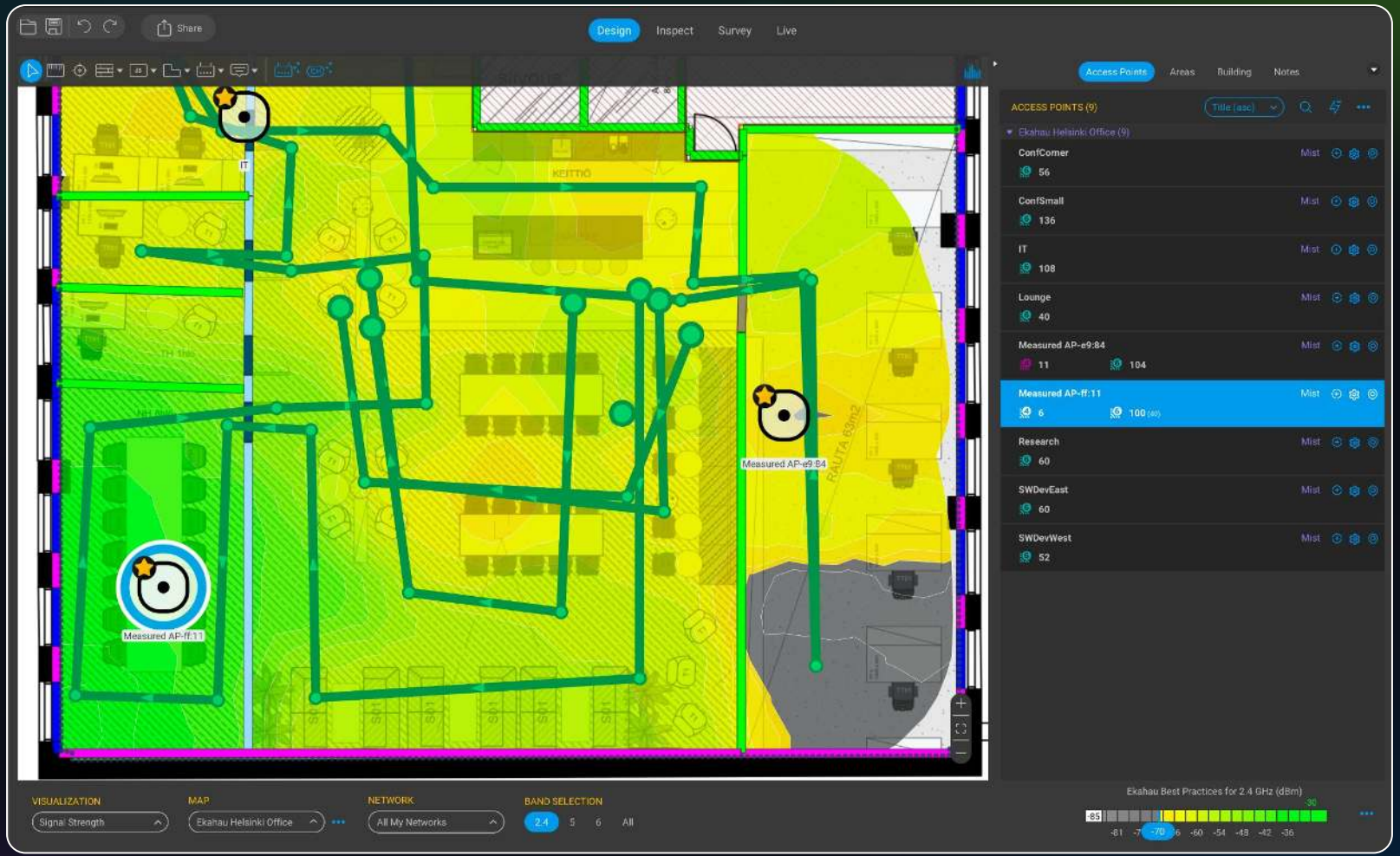

ekahau
High CCI 5GHz Office
Measured environment
Actions
AI Auto-Planner
All done
Optimizing channels
Saving results
63 devices have been optimized to improve Wi-Fi coverage and performance.
Finish
Floor
Floor 3
Band
Wi-Fi 5 GHz
SSIDs
2 SSIDs
Visualization
Channel interference
100 %
radios
1
2
3
4
5
6
3D

OOKLA

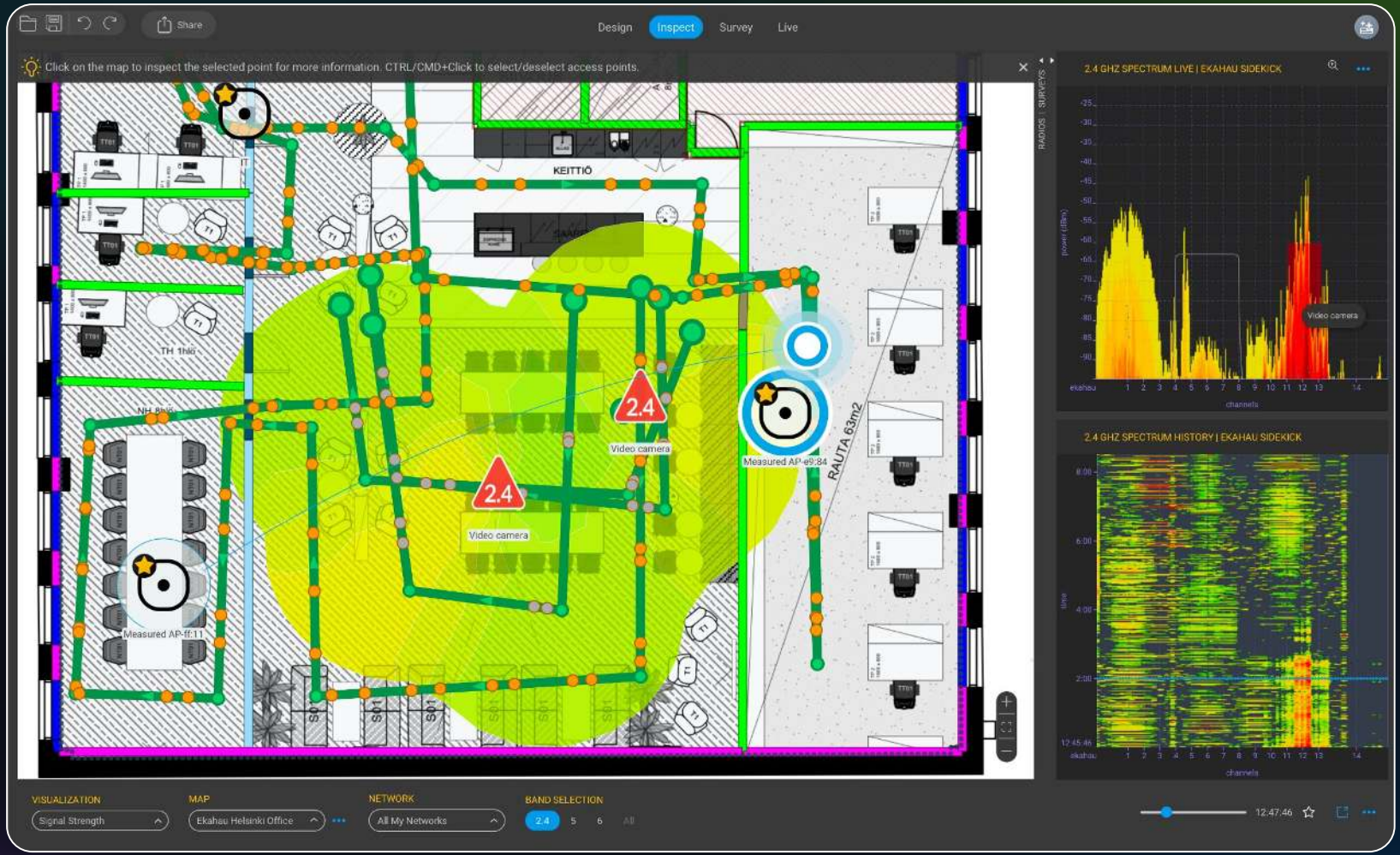

ekahau
High CCI 5GHz Office
Measured environment
Actions
Inspect a location
Signal strength -49 dBm
Secondary signal strength -51 dBm
Signal-to-Noise Ratio (SNR) 46 dB
Channel interference 1 radios
Data rate 244 Mbps
Throughput 212 Mbps
Number of devices 39 devices
Wi-Fi 5 GHz radios
RSSI (dBm)
-49 dBm
5.9 m
Primary
Floor
Floor 3
Band
Wi-Fi 5 GHz
SSIDs
2 SSIDs
Visualization
Channel interference
100 %
radios
1 2 3 4 5 6
3D


“My coffee machine keeps dropping off the Wi-Fi!”


Share
Design
Inspect
Survey
Live
Access Points
Areas
Building
Notes
ACCESS POINTS (9)
Title (asc)
Ekahau Helsinki Office (9)
ConfCorner
Mist
56
ConfSmall
Mist
136
IT
Mist
108
Lounge
Mist
40
Measured AP-e9:84
Mist
11
104
Measured AP-ff:11
Mist
6
100
Research
Mist
60
SWDevEast
Mist
60
SWDevWest
Mist
52
KEITTIÖ
TH 1hl6
NH 8hl6
Measured AP-ff:11
Measured AP-e9:84
RAUTA 63m2
VISUALIZATION
Signal Strength
MAP
Ekahau Helsinki Office
NETWORK
All My Networks
BAND SELECTION
2.4
5
6
All
Ekahau Best Practices for 2.4 GHz (dBm)
-30
-85
-81
-75
-67
-60
-54
-48
-42
-36

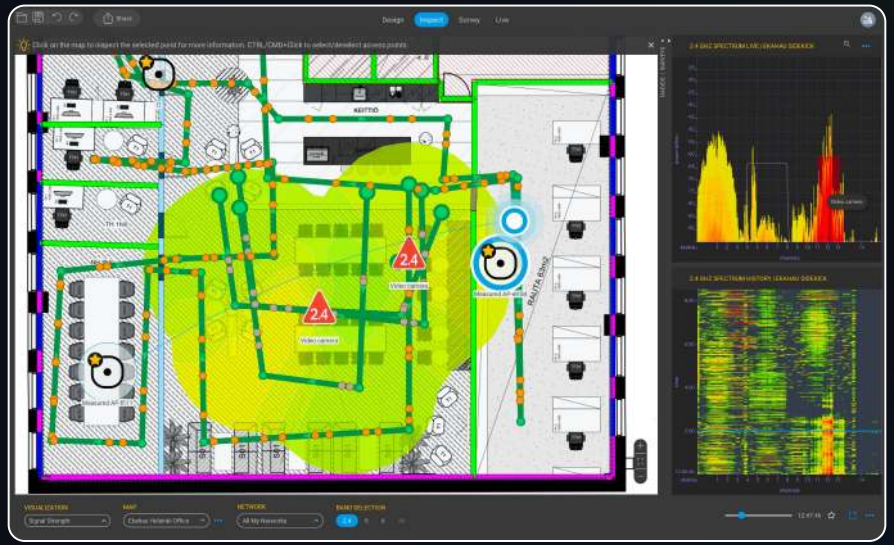

Share
Design
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Access Points
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Building
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ACCESS POINTS (9)
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Ekahau Helsinki Office (9)
ConfCorner
Mist
56
ConfSmall
Mist
136
IT
Mist
108
Lounge
Mist
40
Measured AP-e9:84
Mist
11
104
Measured AP-ff:11
Mist
6
100
Research
Mist
60
SWDevEast
Mist
60
SWDevWest
Mist
52
KEITTIÖ
Measured AP-e9:84
Measured AP-ff:11
VISUALIZATION
Signal Strength
MAP
Ekahau Helsinki Office
NETWORK
All My Networks
BAND SELECTION
2.4
5
6
All
Ekahau Best Practices for 2.4 GHz (dBm)


Share
Design
Inspect
Survey
Live
Access Points
Areas
Building
Notes
ACCESS POINTS (9)
Title (asc)
Ekahau Helsinki Office (9)
ConfCorner
Mist
56
ConfSmall
Mist
136
IT
Mist
108
Lounge
Mist
40
Measured AP-e9:84
Mist
11
104
Measured AP-ff:11
Mist
6
100
Research
Mist
60
SWDevEast
Mist
60
SWDevWest
Mist
52
KEITTIÖ
TH 1hl6
NH 8hl6
Measured AP-ff:11
Measured AP-e9:84
RAUTA 63m2
VISUALIZATION
Signal Strength
MAP
Ekahau Helsinki Office
NETWORK
All My Networks
BAND SELECTION
2.4
5
6
All
Ekahau Best Practices for 2.4 GHz (dBm)
-85
-81
-75
-67
-60
-54
-48
-42
-36
-30


Share
Design
Inspect
Survey
Live
Click on the map to inspect the selected point for more information. CTRL/CMD+Click to select/deselect access points.
KEITTIÖ
TH 1hlö
NH 8hlö
2.4
Video camera
2.4
Video camera
Measured AP-e9:84
RAUTA 63m2
Measured AP-ff:11
2.4 GHZ SPECTRUM LIVE | EKAHAU SIDEKICK
Video camera
2.4 GHZ SPECTRUM HISTORY | EKAHAU SIDEKICK
VISUALIZATION
Signal Strength
MAP
Ekahau Helsinki Office
NETWORK
All My Networks
BAND SELECTION
2.4
5
6
All
12:47:46



Without Spec An



With Spec An

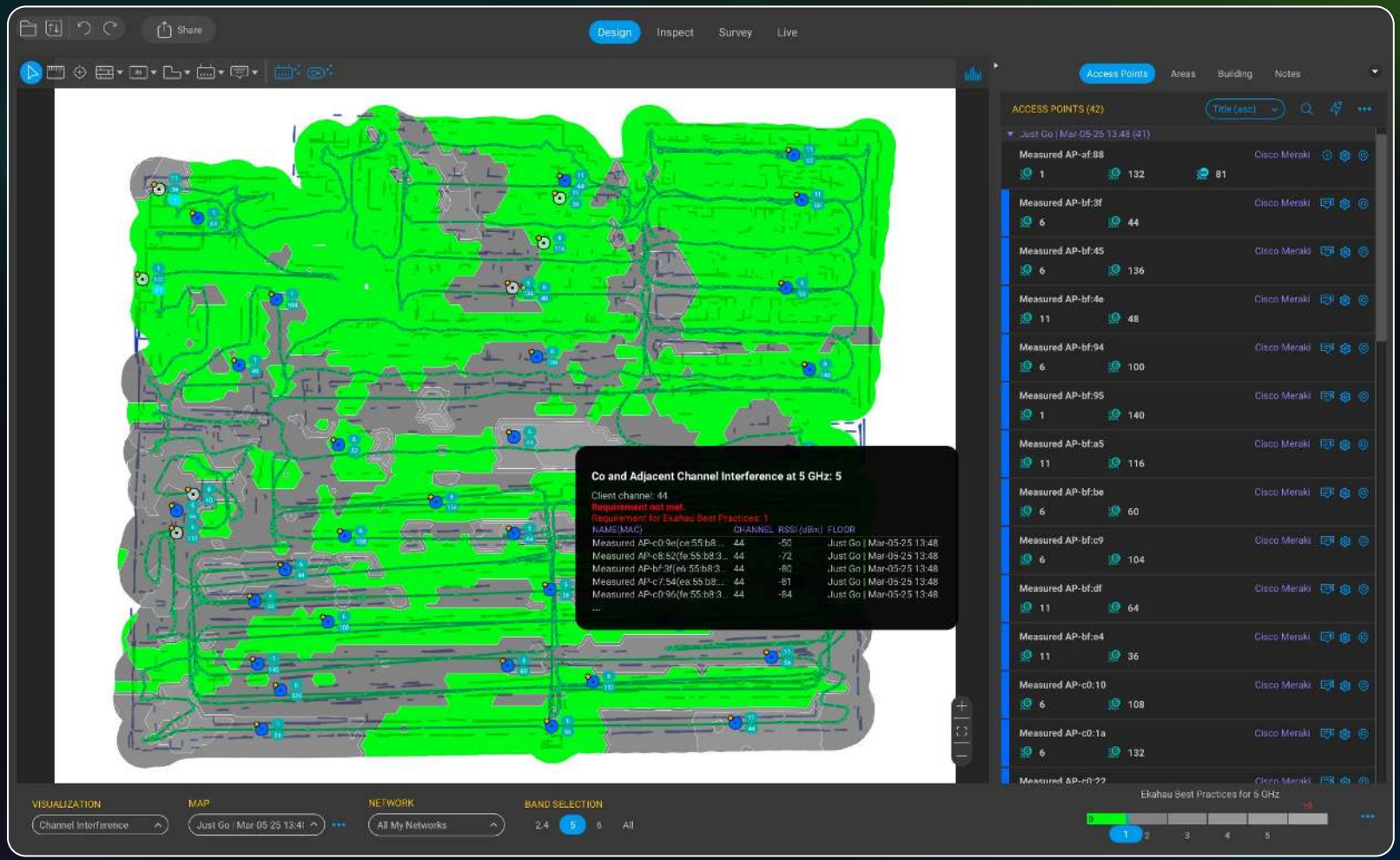
OOKLA



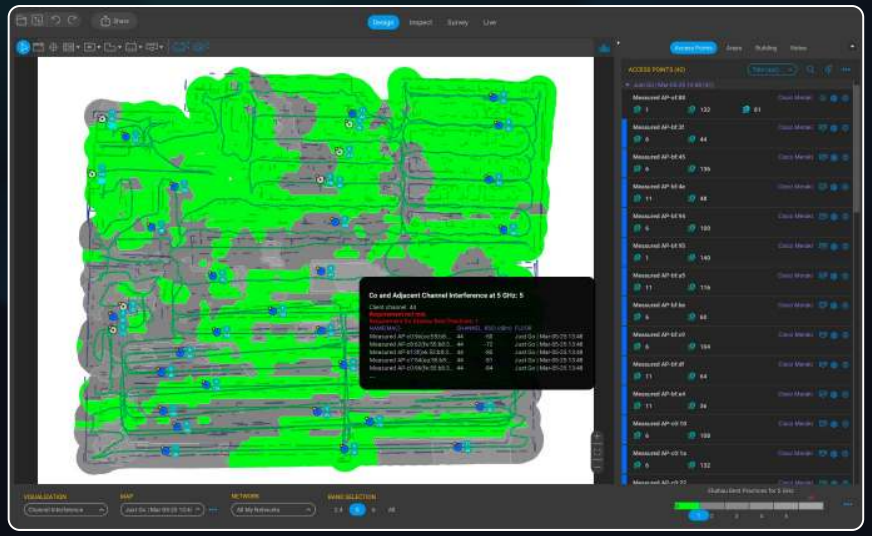
Customer Experiencing Intermittent Wi-Fi Performance

Warehouse Example

OOKLA


Share
Design
Inspect
Survey
Live
Access Points
Areas
Building
Notes
ACCESS POINTS (42)
Title (asc)
Just Go | Mar-05-25 13:48 (41)
Measured AP-af:88
Cisco Meraki
1
132
81
Measured AP-bf:3f
6
44
Measured AP-bf:45
6
136
Measured AP-bf:4e
11
48
Measured AP-bf:94
6
100
Measured AP-bf:95
1
140
Measured AP-bf:a5
11
116
Measured AP-bf:be
6
60
Measured AP-bf:c9
6
104
Measured AP-bf:df
11
64
Measured AP-bf:e4
11
36
Measured AP-c0:10
6
108
Measured AP-c0:1a
6
132
Co and Adjacent Channel Interference at 5 GHz: 5
Client channel: 44
Requirement not met.
Requirement for Ekahau Best Practices: 1
NAME(MAC)
CHANNEL
RSSI (dBm)
FLOOR
Measured AP-c0:9e(ce:55:b8:...
44
-50
Just Go | Mar-05-25 13:48
Measured AP-c8:62(fe:55:b8:3...
44
-72
Just Go | Mar-05-25 13:48
Measured AP-bf:3f(e6:55:b8:3...
44
-80
Just Go | Mar-05-25 13:48
Measured AP-c7:54(ea:55:b8:...
44
-81
Just Go | Mar-05-25 13:48
Measured AP-c0:96(fe:55:b8:3...
44
-84
Just Go | Mar-05-25 13:48
VISUALIZATION
Channel Interference
MAP
Just Go | Mar-05-25 13:48
NETWORK
All My Networks
BAND SELECTION
2.4
5
6
All
Ekahau Best Practices for 5 GHz


Share
Design
Inspect
Survey
Live
Click on the map to inspect the selected point for more information. CTRL/CMD+Click to select/deselect access points.
40
6
56
6
112
5
Continuous transmitter
6
44
6
60
5 GHZ SPECTRUM LIVE | EKAHAU SIDEKICK
power (dBm)
channels
5 GHZ SPECTRUM HISTORY | EKAHAU SIDEKICK
time
channels
VISUALIZATION
Channel Interference
MAP
Just Go | Mar-05-25 13:41
NETWORK
All My Networks
BAND SELECTION
2.4
5
6
All
13:48:21



Without Spec An



With Spec An

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“Can't the Access Points See It?”

Some APs detect interference. Detecting is not the same as identifying, or finding.

A TYPICAL ACCESS POINT

Most have no spectrum analyzer at all

Interference = energy above a threshold

No idea what the interferer actually is

The few that classify can't put it on a map

EKAHAU + SIDEKICK 2

Dedicated wideband spectrum analyzer

Detects and classifies the interference

Places it on the map as you walk

So you can go find it and remove it

Detection tells you something is wrong. Location lets you fix it.

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So to Summarise...

Sub-Optimal Decisions

APs, RRM, and even humans do not always make the best decision when acting blindly without full environment visibility.

The Spectrum Solution

You only truly know what is there and why APs make these decisions if you use a device equipped with a spectrum analyzer.

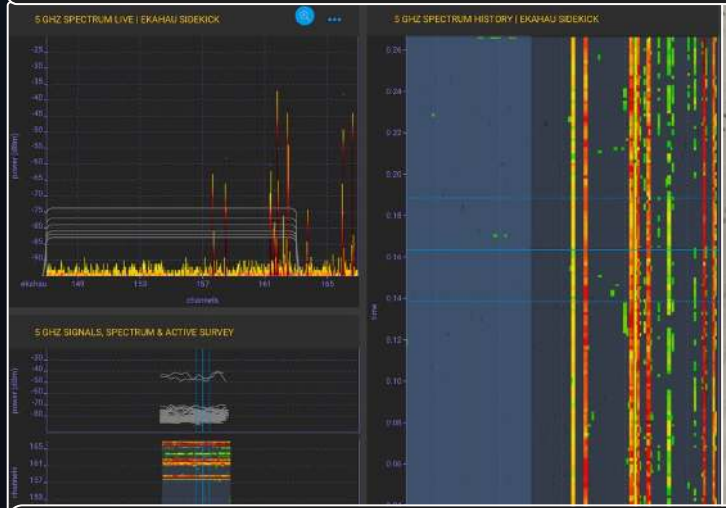
Proprietary & Confidential

Friday • 26/09/2025



Jim Florwick (@cisco.com) 26/09/2025, 15:34

HI Matt. Hope all has been well. I wanted to let you know we just tracked down some monster interference all on Ekahau SK2, I was very impressed with the tools we had and it made short work of a bad problem to get solved (the usual human emotions and career concerns where in full force). Anyway - good story.



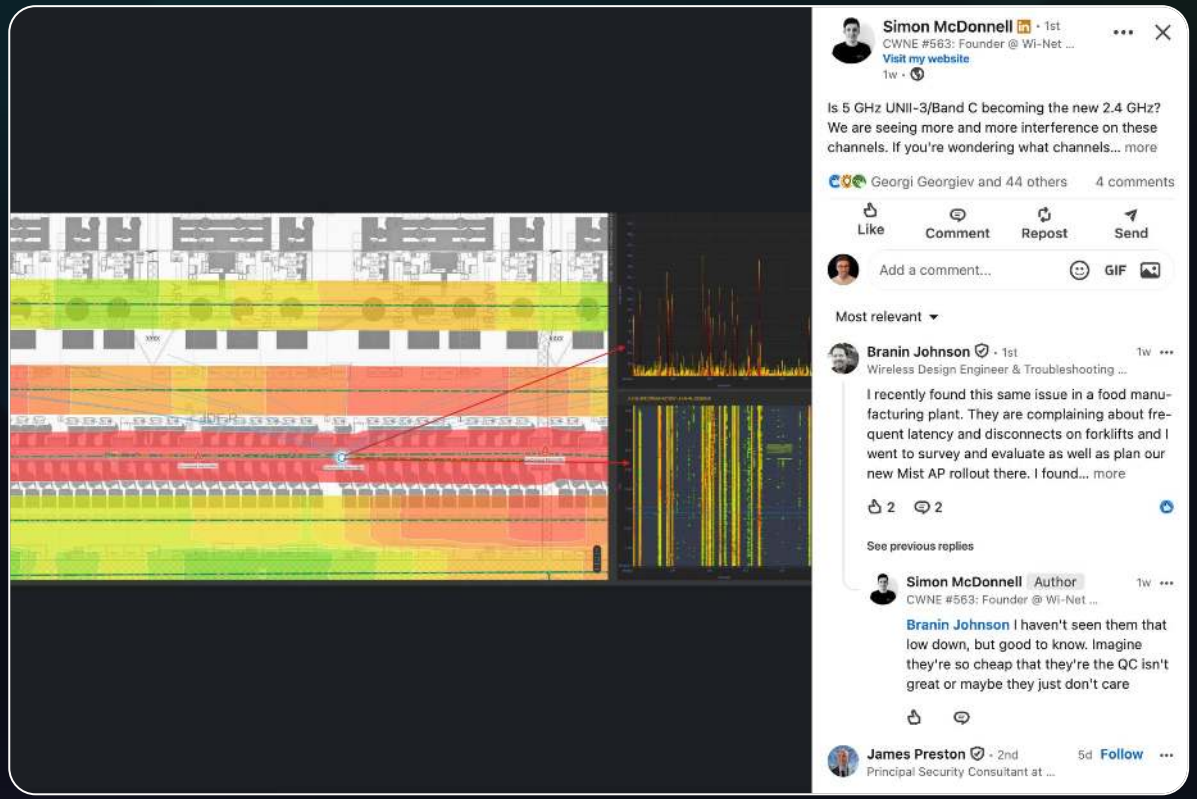




Jim Florwick (@cisco.com) 02/10/2025, 21:22

That above spectrum is multiples of the sensor. I used inspector and moving between samples to see if it was frequency hoping - seems to be just pulsing on frequency, not hoping.

OOKLA



Simon McDonnell • 1st
CWNE #563: Founder @ Wi-Net ...
Visit my website
1w •

Is 5 GHz UNII-3/Band C becoming the new 2.4 GHz? We are seeing more and more interference on these channels. If you're wondering what channels... more

Georgi Georgiev and 44 others 4 comments

Like Comment Repost Send

Add a comment... GIF

Most relevant

Branin Johnson • 1st 1w
Wireless Design Engineer & Troubleshooting ...

I recently found this same issue in a food manufacturing plant. They are complaining about frequent latency and disconnects on forklifts and I went to survey and evaluate as well as plan our new Mist AP rollout there. I found... more

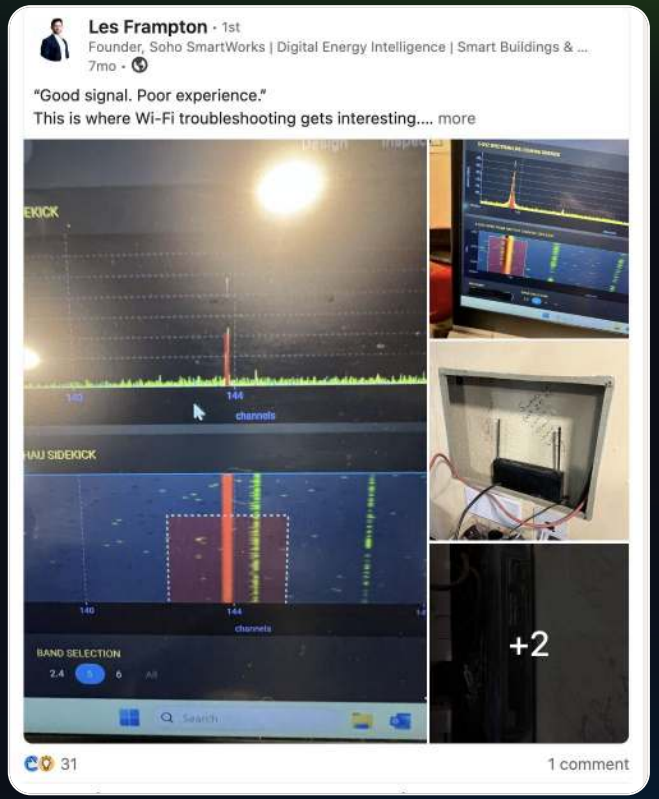
2 2

See previous replies

Simon McDonnell Author 1w
CWNE #563: Founder @ Wi-Net ...

Branin Johnson I haven't seen them that low down, but good to know. Imagine they're so cheap that they're the QC isn't great or maybe they just don't care

James Preston • 2nd 5d Follow
Principal Security Consultant at ...



Les Frampton • 1st
Founder, Soho SmartWorks | Digital Energy Intelligence | Smart Buildings & ...
7mo •

"Good signal. Poor experience."
This is where Wi-Fi troubleshooting gets interesting.... more

31 1 comment

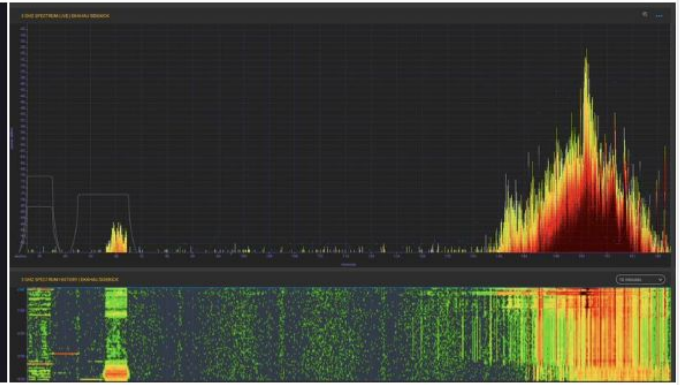
OOKLA

François Vergès · 1st
Wi-Fi Engineer & Founder @ SemFio Networks | Wi-Fi design, sit...
2w ·

UNII-3 is getting busier and busier and not because of Wi-Fi.

On a recent manufacturing site, the customer had reserved the ...more





You and 80 others 36 comments · 3 reposts

Like Comment Repost Send

Ashley Mead · 1st
CWNE #542 | CCNP-W | Wireless Solutions Architect at Natilik
10h ·

I'm definitely in the Spec An Fan camp. I very much enjoy finding an odd signature and then tracking it down. There is so much weirdness out there; I really like to see it.

This captured below has got to be the most severe interference I have ever seen. The signal appeared to be focused on channels 6 and 11, so it masqueraded as Wi-Fi. Of course, when compared to the genuine Wi-Fi signal on channel 1, it wasn't fooling anyone.

Of course, the fun part was tracking the source down. The first screenshot was when it was first detected; the second was when the source was identified.

The source was a scanning machine that used 2.4 GHz to detect the product, then a much smaller wavelength to identify imperfections.

What's the weirdest thing you have found with a Spec An?

#Spectrumanalysis
#Ekahau
#WiFi



You and 14 others 1 comment

“Channel Utilization is one of the most important metrics about Wi-Fi networks and almost always the first thing I check”

So that got me wondering...

How do APs calculate channel utilization?

OOKLA

4

VENDORS ASKED

THE INVESTIGATION

So I asked my friends at 4 of the major AP vendors how they do it...

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Four Different Responses...

1

VENDOR RESPONSE

2

VENDORS RESPONSE

3

VENDORS RESPONSE

4

VENDORS RESPONSE

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Time Spent Talking

The AP's own transmit time, precisely tallied from hardware counters.

Time Spent Listening

Decodable Wi-Fi receive time across its own network and neighboring networks on the same channel.

Blocked by "Something"

Undecodable energy on the channel (non-Wi-Fi), wait states, and NAV timers. The AP knows it's busy, but not why.

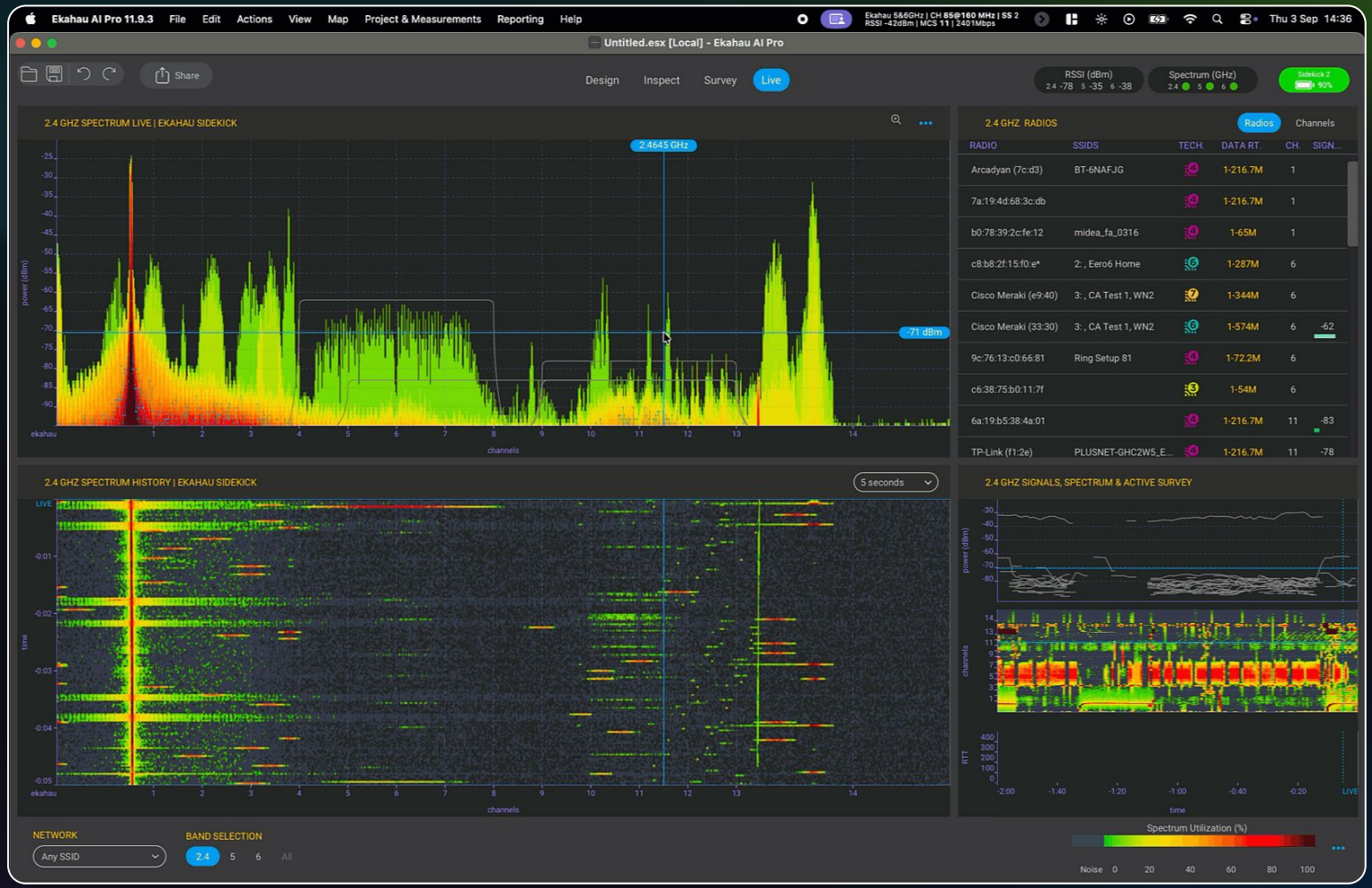
Assumed, Not Measured

Sits on an estimated noise floor varying by chipset, gain, CCA, and firmware. Two APs in one room can report entirely different numbers.

Note: These calculations can vary significantly based on different hardware (HW) and firmware (FW) versions.

Every vendor slices the time differently - but none of them are measuring the RF. If you really want to know your channel utilization, you need a spectrum analyzer.

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Ekahau AI Pro 11.9.3 File Edit Actions View Map Project & Measurements Reporting Help
Thu 3 Sep 14:36
Untitled.esx [Local] - Ekahau AI Pro
Share
Design Inspect Survey Live
RSSI (dBm)
Spectrum (GHz)
2.4 GHZ SPECTRUM LIVE | EKAHAU SIDEKICK
2.4645 GHz
-71 dBm
power (dBm)
channels
2.4 GHZ RADIOS
Radios Channels
RADIO SSIDS TECH. DATA RT. CH. SIGN...
Arcadyan (7c:d3) BT-6NAFJG 1-216.7M 1
7a:19:4d:68:3c:db 1-216.7M 1
b0:78:39:2c:fe:12 midea_fa_0316 1-65M 1
c8:b8:2f:15:f0:e* 2: , Eero6 Home 1-287M 6
Cisco Meraki (e9:40) 3: , CA Test 1, WN2 1-344M 6
Cisco Meraki (33:30) 3: , CA Test 1, WN2 1-574M 6 -62
9c:76:13:c0:66:81 Ring Setup 81 1-72.2M 6
c6:38:75:b0:11:7f 1-54M 6
6a:19:b5:38:4a:01 1-216.7M 11 -83
TP-Link (f1:2e) PLUSNET-GHC2W5_E... 1-216.7M 11 -78
2.4 GHZ SPECTRUM HISTORY | EKAHAU SIDEKICK
5 seconds
time
channels
2.4 GHZ SIGNALS, SPECTRUM & ACTIVE SURVEY
power (dBm)
RTT
NETWORK
Any SSID
BAND SELECTION
2.4 5 6 All
Spectrum Utilization (%)
Noise 0 20 40 60 80 100

Statement

“I don’t always need a spectrum analyzer when I do surveys”

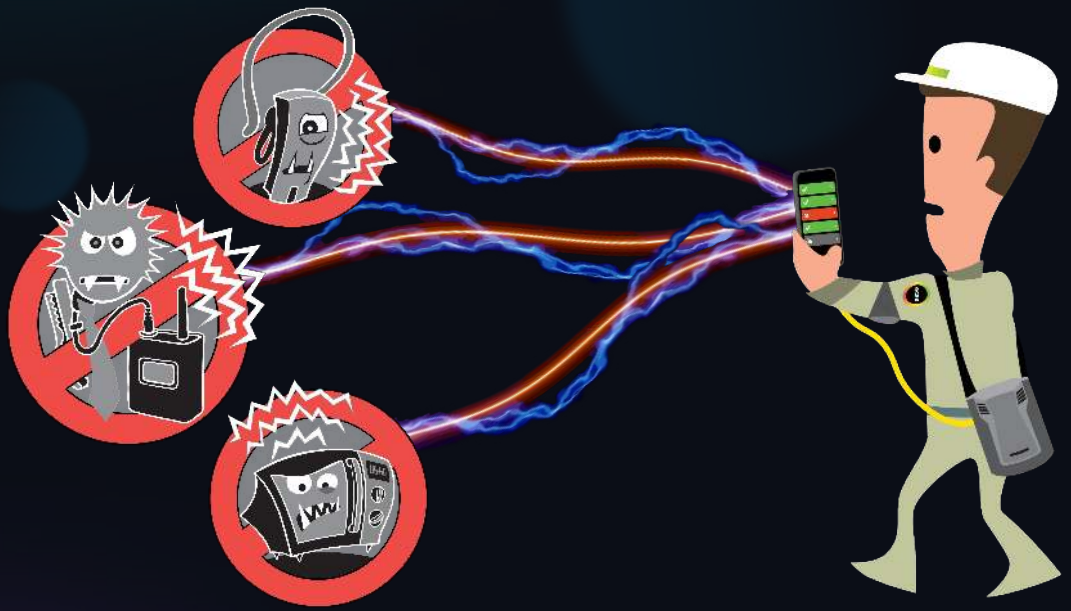
Question

When do you know that you don’t need a spectrum analyzer?

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There are a lot of devices out there

That interfere with your Wi-Fi network that you cannot see

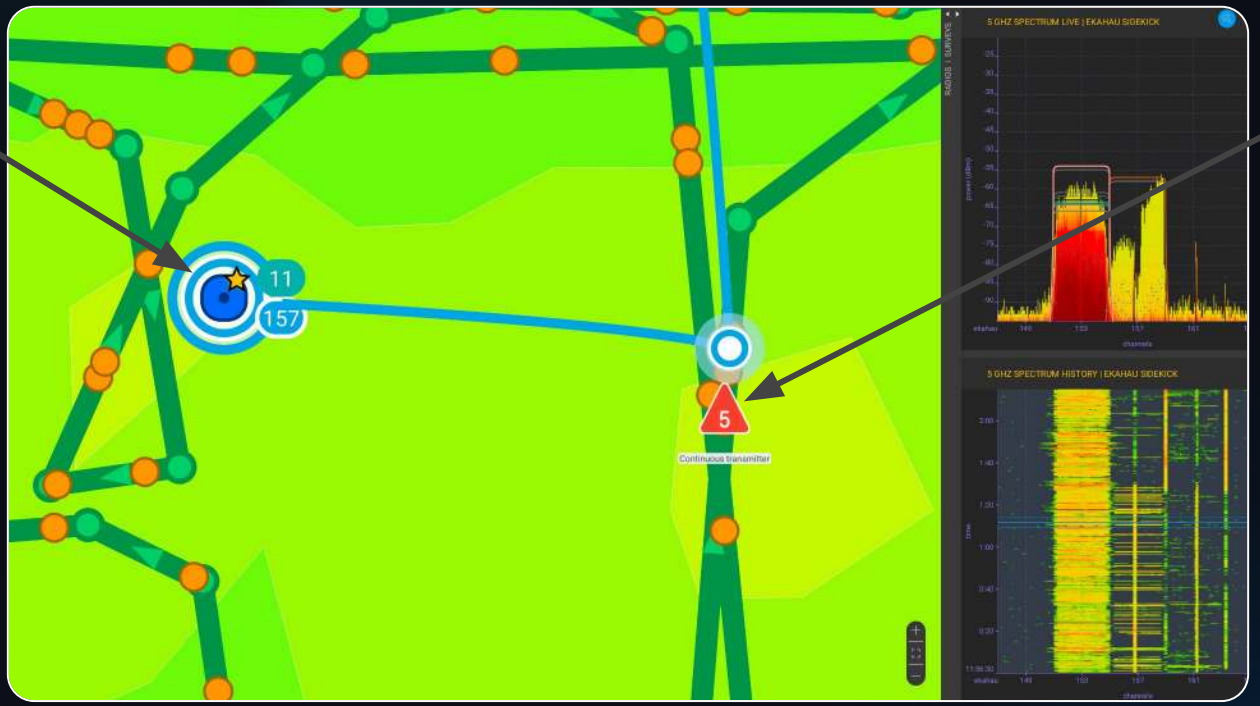




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Ekahau will automatically detect and place

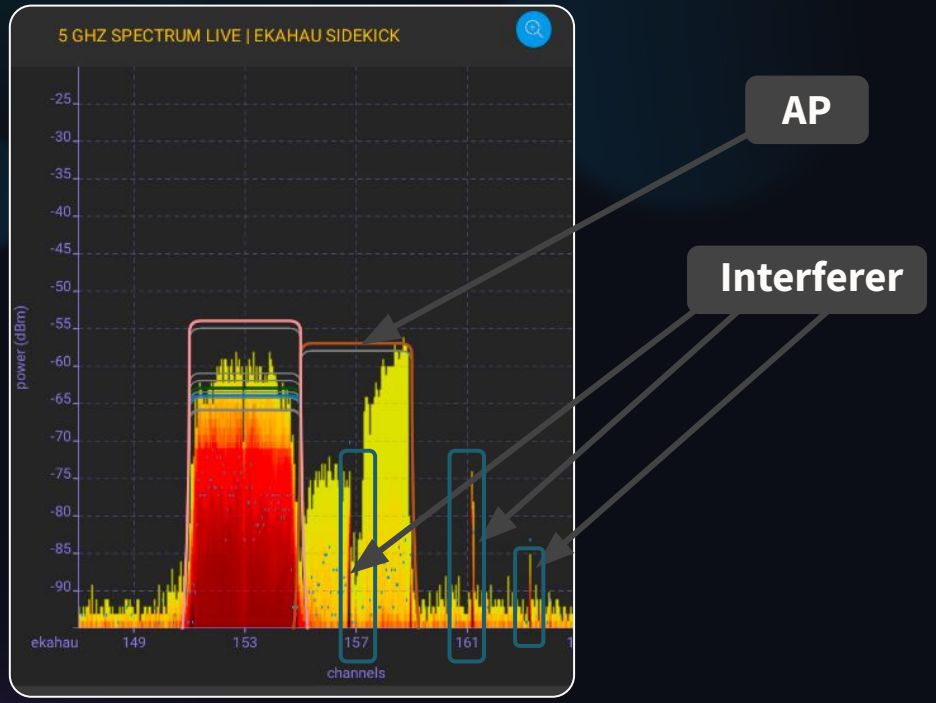
APs and Interferers on your map 📍



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This is important because interferers

Can cause intermittent disruption for your Wi-Fi client devices 📱

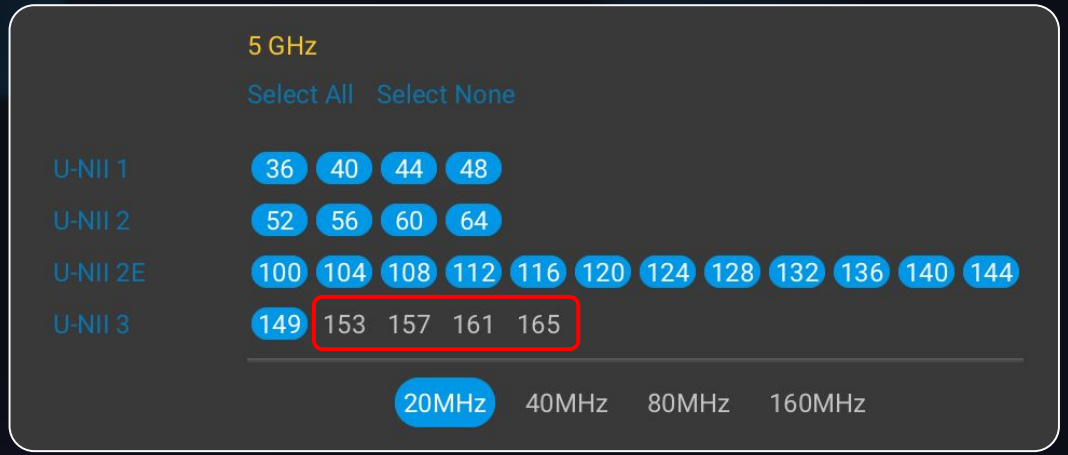


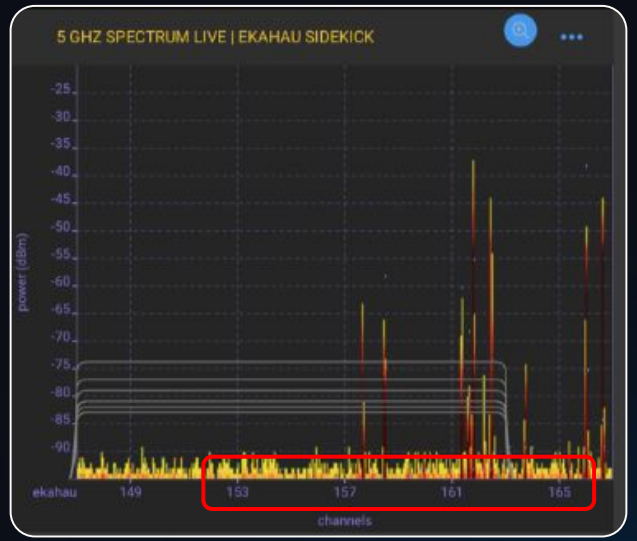
When these devices exist in your network, they can cause your APs and client devices to stop communicating with each other

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Which may make you want to

Remove these interferes or remove affected channels from your channel plan

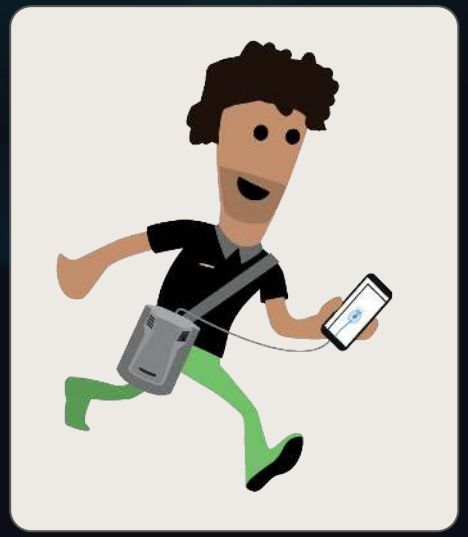


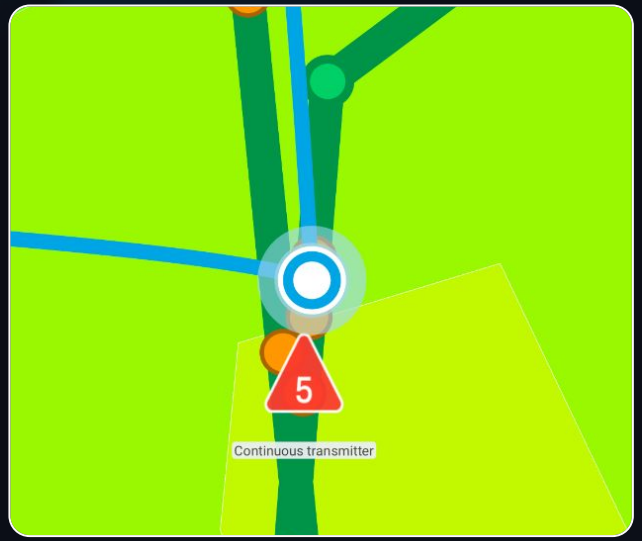


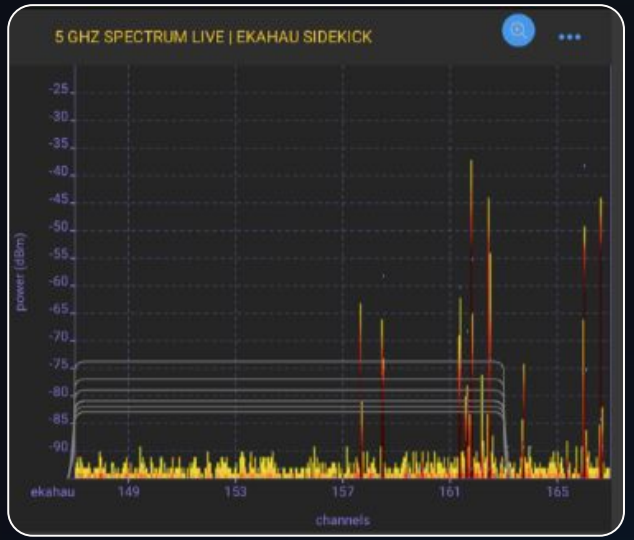
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Performing a survey with Ekahau

Will automatically find and detail the location of these devices 📍



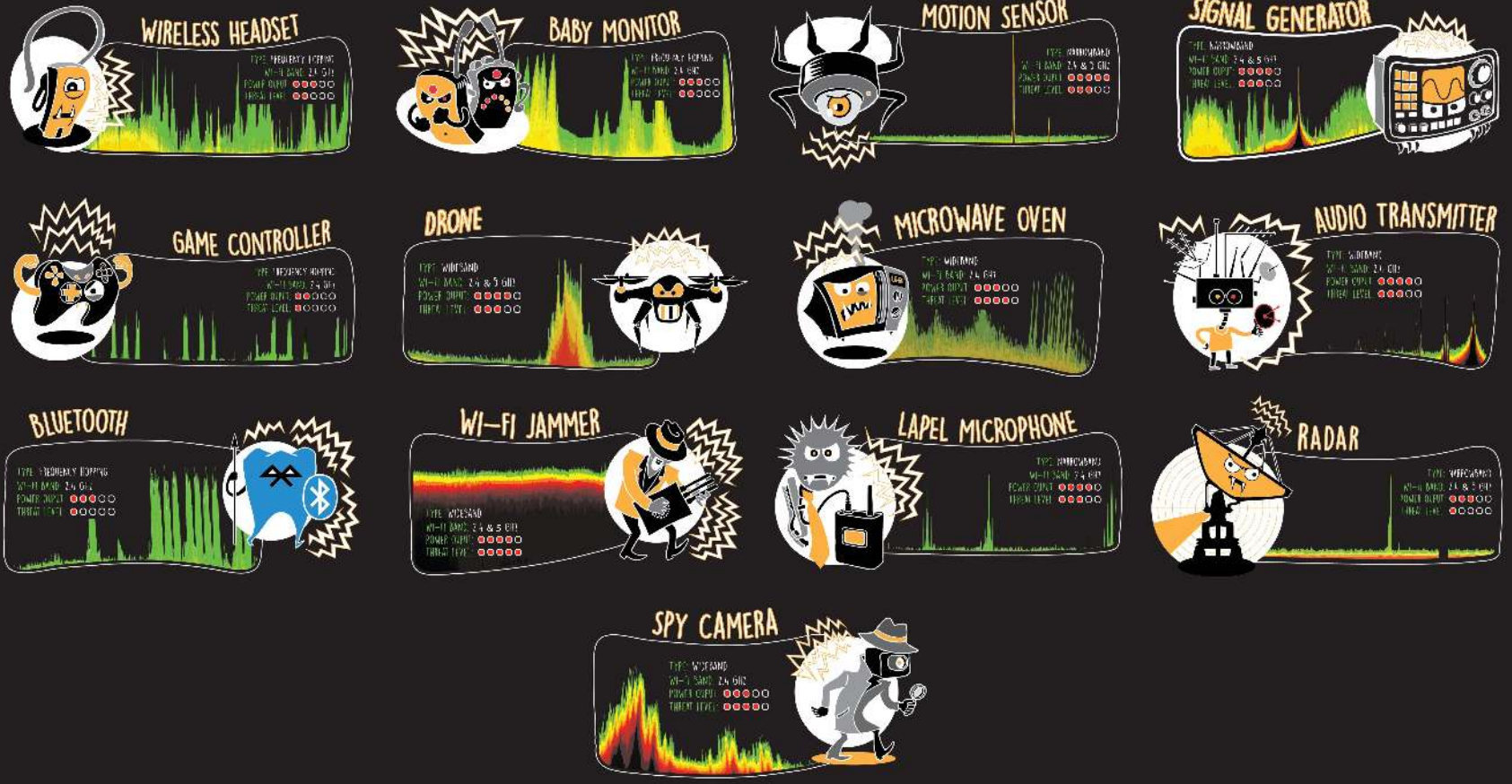




Survey | **Locate** 📍 | **Remove**

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Type of RF Interferers



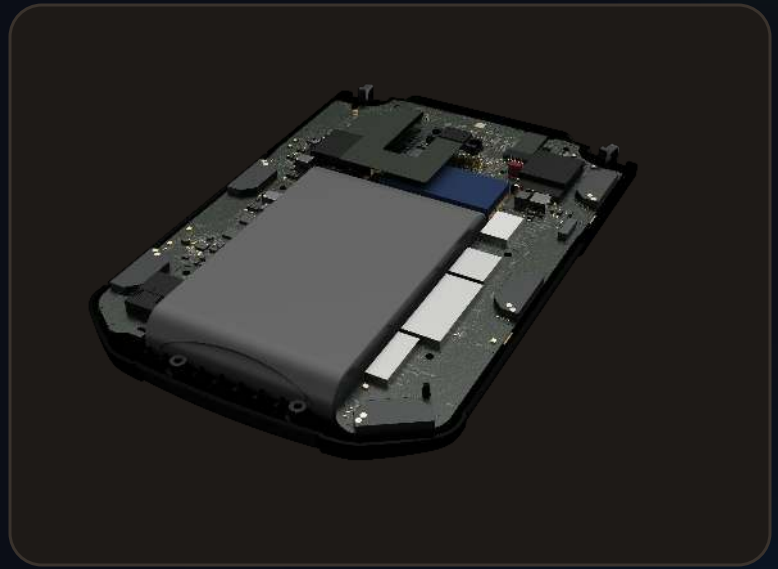
Let me show you something cool



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Spectrum Analyzer

You might not look at it every time, but it’s nice to know it’s there when you need it.



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There will always be cheaper alternatives...





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dBm
-25
-35
-45
-55
-65
-75
-85
-95
2.4 GHz
5 GHz
6 GHz
2.4 GHz
5 GHz
6 GHz
ekahau

I  

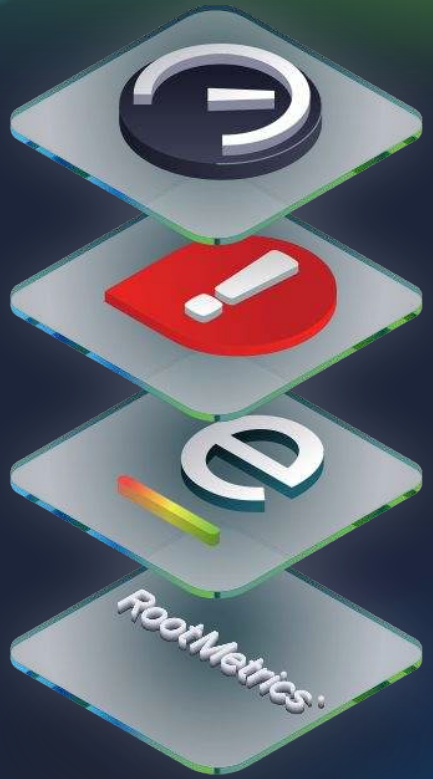
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Oh one more thing...

THANK YOU

Do you have any questions?

ookla.com



SPEEDTEST

Crowdsourced network performance, quality, and availability

Downdetector

Service monitoring, customer engagement, and disruption management

ekahau

Wi-Fi network design, optimization, troubleshooting, and security

RootMetrics

Controlled mobile network testing and benchmarking

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